

Supplementary Material

Evaluation geometry and the limits of cross-region transfer in pre-fire thermal wildfire prediction

Supplementary material for *Evaluation geometry and the limits of cross-region transfer in pre-fire thermal wildfire prediction*.

Section S1 gives the sensitivity analyses and the details behind the Results. Section S2 gives the supporting tables, Section S3 the protocol details and limitations, Section S5 the target-label recovery curve, Section S6 the code and Section S7 the data. Tables are numbered S1 to S34, independently of the sections. Supplementary equations are numbered (S1), (S2) and so on. Every value is read from a released output, named by its path in the repository <https://github.com/dryuemco/thermal-twin>. Paths that begin with `paper/labelfix_rerun/` hold the corrected-label outputs. Values computed under the original Manavgat label are marked “original label”.

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43 The section numbers S1.19 and S4 are not used.

44 **S1 Sensitivity analyses and detail**

45 Each analysis below changes one design choice and keeps everything else fixed. None
 46 changes a conclusion of the main text. Section S1.6 is different: it tests a claim of the
 47 introduction, and the claim is not supported.

S1.1 Evia AOI and prevalence

The raw transfer analyses between Evia and Bejís, Manavgat and Muğla were repeated with the earlier, high-prevalence Evia rectangle. No qualitative conclusion changed. Thermal raw transfer AUCs moved by up to 0.134. The largest change was Evia to Manavgat, at 0.543 with the earlier rectangle against 0.677 with the extended one. No direction changed side of the chance line (paper/labelfix_rerun/round7/r7f_evia_legacy.csv). On the earlier rectangle the natural-vegetation population is smaller, and its burned prevalence is higher than on the extended rectangle (0.287; Table S15).

S1.2 CORAL regularisation

Nine λ values from 0 to 10^{-1} were tested on four directions (Bejís and Muğla, and Manavgat and Muğla, both ways). CORAL transfer AUC moved by at most 0.014 within any direction and model, and by at most 0.009 for the thermal model (paper/labelfix_rerun/exports/coral_lambda_sensitivity_b74d643e/metrics.csv). λ was not selected on performance.

The Manavgat and Bejís pair is not in this sweep. Under the original label, the adapted result of this pair depended on λ . It was therefore tested separately at $\lambda = 10^{-5}$, 10^{-3} and 10^{-1} (paper/labelfix_rerun/step10/coral_lambda_sensitivity.csv). Bejís to Manavgat gives 0.408 [0.389, 0.427], 0.410 and 0.367, which are below chance at every λ . Manavgat to Bejís gives 0.470 [0.444, 0.494], 0.474 and 0.466. No adapted direction of this pair has an interval above chance at any λ , so no conclusion depends on the choice of λ in this range.

S1.3 Blocking scale

The transfer intervals were recomputed at 10-cell (≈ 5 km) blocking from the re-frozen per-cell predictions. The intervals became wider. The paired-delta verdicts changed from twelve

positive, seven negative and one uncertain at 1 km to six, five and nine at 5 km. Coarser blocking therefore removed support from eight verdicts and added none. Across five bootstrap seeds, the 5 km counts were 5 to 6 positive, 4 to 5 negative and 9 to 11 uncertain, so they are not exact (paper/labelfix_rerun/round5/out_official/transfer_ci_blocksize.csv, paper/labelfix_rerun/round6/seed_stability.json).

The point estimates do not change, because the block size only sets the bootstrap resampling unit. Each point estimate is computed once over all target cells. Every verdict that changed moved to “no verdict”, and no direction crossed the chance line. The counts of supported directions are therefore the least stable numbers in this paper.

Table 1 note (resampling units). The bootstrap resamples spatial blocks, so the reliability of an interval depends on the number of blocks with at least one burned cell. This number falls from 192 to 843 at 2 cells to **6 to 33** at 20 cells. An interval built on six such blocks has no useful coverage, so **the 20-cell row should be read as indicative**. The 10-cell row is the coarsest blocking that this design supports. There, every region has 16 to 70 blocks with burned cells, and the gain holds in all five regions.

S1.4 Predictor-window closure

The predictor window was closed 7 and 14 days earlier in all five regions. The thermal contribution stayed positive, with bootstrap support, everywhere. The direction of the change differed between regions. In Manavgat, on the corrected label, it was +0.065 [+0.059, +0.072] at the standard window, +0.082 [+0.075, +0.090] at 7 days and +0.058 [+0.051, +0.065] at 14 days, all on the common cohort of that analysis (paper/labelfix_rerun/exports/window_closure_manavgat_2021/). It became stronger in Bejís, from 0.058 to 0.079 at 14 days, and in Muğla, from 0.115 to 0.128. It was flat in Montiferru. It became weaker in Evia, from 0.156 to 0.149 to 0.135. The four regions other

than Manavgat are not affected by the label correction (paper/labelfix_rerun/exports/window_closure_<region>/tables/thermal_contributions.csv). The gain therefore survives in every region, but it does not always improve.

S1.5 Quality screening of the coarse thermal input

The MODIS inputs were quality-screened in two regions and not in the other three. This split follows the export date, not the design. It causes a change at the input that is correlated with elevation ($r = +0.615$ in Manavgat). The Manavgat chain was therefore rebuilt from a quality-screened input on the corrected label and compared with the unscreened version. The current pipeline step7 refuses the unscreened raster, because the raster has no nodata tag and 8.1 % exact zeros. The unscreened version therefore uses the step7 of export time, and the screened version uses the current one. The two versions differ in code as well as in screening. The result did not change. Elevation stayed at 0.232, no other signed AUC moved by more than 0.005 (downscaled LST, -0.0044), and the within-region gain moved from $+0.067$ to $+0.068$ (paper/labelfix_rerun/round6/qc/qc_compare_manavgat_corrected.json; Section S3.5(xii)). Elevation comes from the DEM and is not affected by the screening. The MODIS-based surface is used for only 2.14 percentage points of coverage. The Muğla input was not re-examined.

S1.6 Normalised against absolute dryness channels

Section 1.2 states that an internally normalised index should be less affected than raw land surface temperature by temperature differences between regions. The thermal set contains both kinds of channels, so this was tested directly by comparing feature sets.

In the two Muğla fires (Section S1.13), the normalised channels keep their direction and the absolute channels do not. This holds for one region across two fires. It does not hold across regions: `1st_anomaly_mean` is one of the predictors with a bootstrap-supported reversal on the

original study areas (Table S10). For this reason, Section S1.14 removes it together with elevation. On this evidence, normalisation protects a channel against a difference between two seasons in one place, but not against a change of place.

Three feature sets were run over all twenty directions and all five within-region folds. The classifier, population, folds and bootstrap were kept fixed. The code stops unless its reference configuration reproduces the re-frozen exports, and it does: the largest absolute difference from the step9b transfer AUCs is 0.000000 across all twenty directions. Sources for this section and the next two: `paper/labelfix_rerun/round3/no_coord_channels.json` and `paper/labelfix_rerun/code/model_capacity.json`, summarised by `paper/labelfix_rerun/round7/r7e_supplement_tables.py`.

Table S23. Feature-set arms. Mean over the twenty transfer directions and the five regions.

Feature set	Mean transfer AUC	Directions > 0.5	Mean within-region AUC
Baseline only	0.519	13	0.797
Baseline + normalised anomalies	0.524	14	0.862
Baseline + absolute surface state	0.531	13	0.864
All ten features (reference)	0.527	13	0.896

The expected advantage was not found. Per direction, the normalised set minus the absolute set has a mean of -0.006 . The difference is positive in 11 of 20 directions and ranges from -0.099 to $+0.087$, so the spread is much larger than the mean difference. Within regions the two sets are almost equal, at 0.862 and 0.864. Each recovers most of the gap between the baseline (0.797) and the full model (0.896). The comparison uses two parts of one thermal set on one cohort. It does not test normalised dryness indices in general.

S1.7 The coordinate-informed channels

`downscaled_1st_mean` and `fused_1st_mean` come from a per-region downscaling model that uses coordinates as inputs (Section S3.4). This is the only way in which coordinate

information enters the feature set. A surface smoothed by coordinates is informative locally and does not transfer. If the within-region gain depended on it, the gain would be caused by the smoothing and not by the thermal state. Both channels were therefore removed, and all models were refitted over all twenty directions and all five within-region folds.

Table S24. The within-region increment without the two coordinate-informed channels.

Region	Increment, full set	Without the two channels	Retained
Manavgat 2021	+0.067	+0.064	95 %
Bejís 2022	+0.056	+0.046	82 %
Muğla 2021	+0.116	+0.097	84 %
North Evia 2021	+0.153	+0.145	94 %
Montiferru 2021	+0.101	+0.105	103 %
Mean	+0.099	+0.091	92 %

The gain does not depend on these channels. It stays positive in every region and keeps 82 % to 103 % of its size. Mean transfer also does not change: 0.528 without the two channels, 0.527 with them, and 0.519 for the baseline alone.

S1.8 Model capacity

All numbers in the paper come from one random forest with unlimited depth and `min_samples_leaf = 3`. This configuration fits local structure well and extrapolates poorly. Three other estimators were therefore run over the same twenty directions and five within-region folds.

Table S25. Four estimators, within region and in transfer.

Estimator	Within-region AUC	Within increment	Transfer AUC	Transfer increment	Above chance
Random forest, depth unlimited, leaf 3	0.896	+0.099	0.527	+0.007	13 of 20
Random forest, depth 6, leaf 50	0.838	+0.053	0.527	−0.011	13 of 20
Random forest, leaf 200	0.814	+0.046	0.522	−0.019	13 of 20
Penalised logistic regression	0.759	+0.046	0.481	−0.021	11 of 20

Regularisation does not improve transfer. The four estimators range from 0.481 to 0.527, with eleven to thirteen of twenty directions above chance. The linear model, which is

designed to extrapolate, transfers worst. Within-region skill falls from 0.896 to 0.759 as capacity is reduced, but transfer does not rise. The thermal contribution to transfer is +0.007 for the standard forest and negative for all three alternatives. The estimators are not ranked, because only point estimates are available.

S1.9 The four evaluations of Section 4.3, in full

Section 4.3 reports four evaluations of the same models (main-text Table 2). The values per scar, the controls and the thermal gain per scar are given here.

Table S1. The four evaluations, per scar. Natural-vegetation population, 5 km blocking, 2 km collar. Bejís and Manavgat have no row C, because each has only one burned area, and withholding it leaves nothing to train on. Source [paper/labelfix_rerun/code/matched_holdout.json](https://github.com/labelfix_rerun/code/matched_holdout.json).

Region	Scar	Cells	Burned fraction	A whole region	B scar area	C scar withheld	D foreign
Manavgat 2021	1	3,965	0.74	0.882	0.743	n/a	0.542
Bejís 2022	1	1,657	0.66	0.824	0.575	n/a	0.589
Muğla 2021	6	1,266	0.72	0.777	0.564	0.595	0.571
Muğla 2021	1	1,244	0.59	0.777	0.654	0.561	0.514
Muğla 2021	10	940	0.68	0.777	0.674	0.617	0.634
Muğla 2021	8	954	0.57	0.777	0.761	0.542	0.623
North Evia 2021	1	3,059	0.87	0.864	0.745	0.465	0.564
Montiferru 2021	1	758	0.58	0.720	0.622	0.584	0.489
Montiferru 2021	5	195	0.34	0.720	0.461	0.458	0.475

S1.9.1 The pool controls

The same predictions were scored on a random sample of region cells with the burned fraction of the scar area. Over nine scars this gives 0.793, against 0.791 for the whole region. Matching the prevalence therefore changes nothing (−0.002 [−0.005, +0.001]). This is expected, because ROC-AUC does not depend on class balance, so it is a check of the code. Scoring the predictions on the scar area gives 0.644, a fall of **+0.149 [+0.087, +0.211]** from

the prevalence-matched score. This replaces both the burned and the unburned cells, so each group was also replaced alone. The cost is always the prevalence-matched score minus the new score. Replacing only the unburned cells costs **0.139 [0.098, 0.181]** (0.046 to 0.250 per scar). Replacing only the burned cells costs **-0.002 [-0.055, +0.051]** on average (-0.121 to +0.137 per scar). The cost therefore comes from the unburned cells. All unburned cells in a scar collar are next to the fire and share its terrain, land cover and weather. The unburned cells of a whole region also include easy distant cells (paper/labelfix_rerun/code/pool_decomposition.json, prevalence_control.json).

S1.9.2 The seven scars and the resampling unit

Rows A to D of Table 2 are means over the **same seven scars**: four in Muğla, two in Montiferru and one in Evia. Under the corrected label, the 2,151 added Manavgat burned cells join its one existing scar (2,934 of 2,935 cells). For this reason the controls above use nine scars and report 0.791 and 0.644, against 0.773 and 0.640 in Table 2. The intervals are Student *t* intervals over the scars. **Row A is a region-level value that is repeated for each scar of a region, so its interval is too narrow and should not be read as coverage.** The same applies to A – B and A – C. With the region as the unit, A – B is **+0.137 [+0.048, +0.226]** over three regions and **+0.160 [+0.090, +0.230]** over all nine scars in five regions (paper/labelfix_rerun/round7/round7_summary.json). A – C with the region as the unit is **+0.266 [-0.022, +0.553]**. This interval is 3.6 times wider than the scar-level interval and includes zero. **With the region as the unit, only the frame cost is established.**

The scar definition does not drive the result. Over a range of minimum scar sizes and buffers, row C ranges from 0.541 to 0.561 on nine to six scars. Its means at 2, 5 and 10 km buffers are 0.546 (seven scars), 0.533 (seven) and 0.548 (six) (paper/labelfix_rerun/code/scar_definition_sweep.csv, scar_control.json). For

single scars, row D varies strongly between sources (Table S4), but little after averaging. Row D uses 28 of the 36 source and scar combinations, and nine of them are below chance.

S1.9.3 Controls and the thermal-increment ladder, full specification

Three controls reuse the predictions of row A. Each is averaged over 20 draws without replacement. The **prevalence-matched** control draws $|H_K^+|$ cells from P_R and $|H_K^-|$ cells from $V_R \setminus P_R$. The **negative-pool** control keeps the drawn burned cells but uses the unburned cells of the scar, H_K^- . The positive-pool control does the opposite. A **within-region half-split** completes the set: the modelled cells are cut at the median of a grid axis, in both axes and both directions, and the transfer protocol of Section 3.8 is applied. A split is dropped when one half has only one class. The numbers of burned cells in the two halves are unequal (Table S2).

The **thermal gain** is evaluated with rows A to D on the seven scars of Table 2. Table S26 gives the values per scar at both block sizes. Over the seven scars, the region-wide gain is +0.095 at 5 km blocking and +0.117 at 1 km. On the scar area, the same predictions give +0.021 and +0.056. Leave-one-scar-out gives +0.024, and a foreign-region model +0.008, at both block sizes. Blocked minus leave-one-scar-out is -0.004 $[-0.070, +0.063]$ at 5 km and $+0.031$ $[-0.027, +0.090]$ at 1 km. Region-wide minus scar area at 5 km is $+0.074$ $[-0.009, +0.157]$ at scar level and $[+0.034, +0.146]$ with a region-cluster bootstrap. The within-region half-split gives +0.028, positive in 13 of 18 splits (paper/labelfix_rerun/inference/ladder_summary.json, paper/labelfix_rerun/code/positive_control.json).

Table S26. The thermal gain by scar: region-wide, on the scar area, withheld and foreign. Thermal minus baseline ROC-AUC. Region-wide and scar-area values at 5 km

219 blocking, with 1 km in brackets. Leave-one-scar-out and foreign values do not depend on the
 220 blocking.

Region	Scar	Region-wide	Scar frame, same blocked model	Scar withheld	Foreign
Muğla 2021	6	+0.079 (+0.116)	−0.016 (+0.018)	+0.025	+0.033
Muğla 2021	1	+0.079 (+0.116)	+0.049 (+0.071)	−0.044	+0.014
Muğla 2021	10	+0.079 (+0.116)	+0.063 (+0.107)	+0.045	+0.042
Muğla 2021	8	+0.079 (+0.116)	+0.083 (+0.092)	+0.012	+0.037
North Evia 2021	1	+0.148 (+0.153)	+0.060 (+0.065)	+0.143	−0.038
Montiferru 2021	1	+0.100 (+0.101)	+0.066 (+0.086)	+0.060	−0.008
Montiferru 2021	5	+0.100 (+0.101)	−0.160 (−0.049)	−0.070	−0.023
Mean		+0.095 (+0.117)	+0.021 (+0.056)	+0.024	+0.008

221 **Table S2. Within-region half-split, every split.** The numbers of burned cells in the source
 222 and target halves are given because they are unequal. This is the main limit of this analysis,
 223 because a straight cut does not give two comparable halves. Two splits cannot be used,
 224 because one half of Manavgat has no burned cells. Source
 225 `paper/labelfix_rerun/code/positive_control.json`; checked row by row by
 226 `paper/code/appendix_tables.py`.

Region	Axis	Direction	Source positives	Target positives	Thermal AUC	Baseline AUC
manavgat 2021	east-west	low to high	1,795	1,140	0.761	0.793
manavgat 2021	east-west	high to low	1,140	1,795	0.712	0.638
manavgat 2021	north-south	low to high	0	2,935	n/a	n/a
manavgat 2021	north-south	high to low	2,935	0	n/a	n/a
bejis 2022	east-west	low to high	640	460	0.750	0.784
bejis 2022	east-west	high to low	460	640	0.578	0.560
bejis 2022	north-south	low to high	357	743	0.713	0.662
bejis 2022	north-south	high to low	743	357	0.603	0.613
mugla 2021	east-west	low to high	1,449	1,462	0.448	0.378
mugla 2021	east-west	high to low	1,462	1,449	0.636	0.628
mugla 2021	north-south	low to high	774	2,137	0.533	0.503

Region	Axis	Direction	Source positives	Target positives	Thermal AUC	Baseline AUC
mugla 2021	north-south	high to low	2,137	774	0.294	0.549
evia 2021 extended	east-west	low to high	2,052	612	0.589	0.506
evia 2021 extended	east-west	high to low	612	2,052	0.645	0.482
evia 2021 extended	north-south	low to high	2,564	100	0.649	0.513
evia 2021 extended	north-south	high to low	100	2,564	0.533	0.495
montiferru 2021	east-west	low to high	410	129	0.475	0.461
montiferru 2021	east-west	high to low	129	410	0.507	0.513
montiferru 2021	north-south	low to high	438	101	0.583	0.511
montiferru 2021	north-south	high to low	101	438	0.536	0.447

Table S3. Leave-one-scar-out at a 2 km buffer, every scar. Only Muğla and Montiferru contain more than one burned area of at least 50 cells. Only in Muğla does every held-out case still leave a well-trained source model. The two Montiferru areas are very unequal, so one case keeps 472 burned source cells and the other 97. The mean is 0.579 for the four Muğla cases and 0.502 for the three others. The pooled mean, 0.546, is row C of Table 2. Source paper/labelfix_rerun/code/scar_control.json; checked row by row by paper/code/appendix_tables.py.

Region	Component	Source positives left	Target positives	Target cells	AUC
evia 2021 extended	1	11	2,653	3,059	0.465
montiferru 2021	1	97	442	758	0.584
montiferru 2021	5	472	67	195	0.458
mugla 2021	6	1,997	914	1,266	0.595
mugla 2021	1	2,173	738	1,244	0.561
mugla 2021	10	2,272	639	940	0.617
mugla 2021	8	2,363	548	954	0.542

The evaluation populations of these analyses cannot be compared with each other or with the transfer targets. A held-out scar with its 2 km collar contains only unburned cells next to the fire, while a whole target region also contains easy distant cells. The burned fractions (34 to 87 %, against 7.0 to 28.7 %) are a result of this difference, not its cause.

Table S4. The foreign-region evaluation, by source. Each held-out scar area is scored with a model fitted on each of the other four regions. Row D of Table 2 is the mean over the seven scars that have a row C (28 of the 36 combinations below). Over all nine scars the mean is 0.556. Source `paper/labelfix_rerun/code/d_per_source.json`; checked row by row by `paper/code/appendix_tables.py`.

Target region	Scar	Mean over sources	Min	Max	Spread
Bejís 2022	1	0.589	0.519	0.631	0.112
North Evia 2021	1	0.564	0.374	0.704	0.329
Manavgat 2021	1	0.542	0.438	0.665	0.227
Montiferru 2021	1	0.489	0.457	0.513	0.056
Montiferru 2021	5	0.475	0.399	0.597	0.197
Muğla 2021	1	0.514	0.410	0.608	0.198
Muğla 2021	6	0.571	0.545	0.597	0.052
Muğla 2021	8	0.623	0.533	0.724	0.192
Muğla 2021	10	0.634	0.433	0.716	0.283

Over all 36 combinations the mean is 0.556, the range is 0.374 to 0.724, and ten combinations are below chance. The mean spread across the four sources for one scar is 0.183 over nine scars and 0.187 over seven. Row D is averaged over sources so that it can be compared with row C, which is fitted on one region. This does not mean that the choice of foreign source is unimportant.

Table S5. Prevalence is not the cause of the evaluation-area effect. The same fitted model and the same out-of-fold predictions are scored in three ways: on the whole region, on a random sample of region cells with the burned fraction of the scar area, and on the scar area. Twenty draws per scar, seed 42. Nine scars are used, because this control needs no leave-one-scar-out model. Values at 4 dp, as stored in the source `paper/labelfix_rerun/code/prevalence_control.json`; checked row by row by `paper/code/appendix_tables.py`.

Target region	Scar	Burned fraction	A whole region	A' prevalence-matched	B scar area
Manavgat 2021	1	0.74	0.8822	0.8837	0.7434
Bejís 2022	1	0.66	0.8245	0.8251	0.5748

Target region	Scar	Burned fraction	A whole region	A' prevalence-matched	B scar area
Muğla 2021	6	0.72	0.7773	0.7782	0.5640
Muğla 2021	1	0.59	0.7773	0.7787	0.6540
Muğla 2021	10	0.68	0.7773	0.7835	0.6735
Muğla 2021	8	0.57	0.7773	0.7796	0.7609
North Evia 2021	1	0.87	0.8642	0.8632	0.7451
Montiferru 2021	1	0.58	0.7199	0.7179	0.6215
Montiferru 2021	5	0.34	0.7199	0.7285	0.4611
Mean			0.7911	0.7932	0.6443

A minus A', the effect of prevalence alone, is -0.002 [-0.005 , $+0.001$]. A' minus B, which replaces both groups of cells, is $+0.149$ [$+0.087$, $+0.211$]. Section S1.9.1 shows that it comes from the unburned cells.

S1.10 The transfer-gap decomposition, in full

Among the directions that started below chance, the recovered fraction of Section 3.10 is at most $+0.28$ (Bejís to Evia). Seven directions have negative recovery, five of them with intervals fully below zero. Manavgat to Muğla (-0.03 [-0.06 , $+0.01$]) and Muğla to Bejís (-0.07 [-0.16 , $+0.01$]) are negative at the point estimate only. Values are from `four_aoi_decomposition.csv` of the re-frozen outputs (`paper/labelfix_rerun/round5/tables/corrected/`). The within-region reference in the denominator is not matched to a transfer evaluation (Section 4.3), and the raw column depends on the study area (Section 4.4). The fractions are therefore only valid within this protocol.

Table S6. Transfer-gap decomposition (four-AOI set, 12 directions). Within = within-region thermal AUC of the target. Best adapted = the better of z-score and CORAL; this choice uses target labels. Recovered fraction = (adapted – raw)/(within – raw), signed and not clipped, with a paired bootstrap CI (1000 replicates, 2-cell blocks). Montiferru directions are not part of this decomposition. The status column shows whether the *adapted* value is above chance, using the 2-cell adapted intervals of Table S16.

Direction	Within	Raw	Best adapted (method)	Recovered fraction [CI]	Status
Bejís→Evia	0.912	0.383	0.532 (z-score)	+0.28 [+0.23, +0.32]	adapted above chance
Muğla→Manavgat	0.908	0.345	0.485 (z-score)	+0.25 [+0.22, +0.28]	adapted, chance not excluded
Evia→Bejís	0.918	0.448	0.549 (z-score)	+0.22 [+0.16, +0.27]	adapted above chance
Bejís→Manavgat	0.908	0.314	0.406 (CORAL)	+0.15 [+0.13, +0.18]	adapted still below chance
Manavgat→Bejís	0.918	0.396	0.467 (CORAL)	+0.14 [+0.09, +0.17]	adapted still below chance
Manavgat→Muğla	0.859	0.438	0.427 (z-score)	–0.03 [–0.06, +0.01]	negative recovery
Muğla→Bejís	0.918	0.583	0.560 (CORAL)	–0.07 [–0.16, +0.01]	negative recovery
Evia→Muğla	0.859	0.577	0.530 (CORAL)	–0.17 [–0.22, –0.11]	negative recovery
Muğla→Evia	0.912	0.653	0.563 (CORAL)	–0.35 [–0.43, –0.27]	negative recovery
Bejís→Muğla	0.859	0.618	0.518 (z-score)	–0.42 [–0.51, –0.34]	negative recovery
Manavgat→Evia	0.912	0.654	0.529 (z-score)	–0.48 [–0.56, –0.42]	negative recovery
Evia→Manavgat	0.908	0.677	0.417 (CORAL)	–1.13 [–1.27, –1.00]	negative recovery

S1.11 The thermal sign, stratified

Section 4.4 reports that four of five regions show a negative association between pre-fire surface temperature and burning, and that Manavgat shows the same once elevation is held. The values per region are given here. Signed AUC against burned within the 10 km collar. The stratified columns pool the concordance within deciles of the named variable, and each decile is weighted by its number of burned and unburned cell pairs. Sources: paper/labelfix_rerun/code/matched_frame_gap.csv (LST raw, within NDVI, within distance; NDVI columns), which can be recomputed by paper/code/verify_matched_gap.py, and paper/labelfix_rerun/round7/r7b_lst_given_terrain.csv (LST within elevation, and LST detrended on elevation).

Table S27. The LST association, stratified, 10 km collar.

Region	LST raw	LST within elevation	LST detrended on elevation	LST within NDVI	LST within distance	NDVI raw	NDVI within LST
Manavgat	0.522	0.403	0.409	0.556	0.454	0.551	0.550
Bejís	0.405	0.509	0.411	0.486	0.350	0.618	0.574
Muğla	0.332	0.363	0.365	0.404	0.367	0.652	0.547
Evia	0.286	0.327	0.312	0.279	0.319	0.663	0.380
Montiferru	0.376	0.392	0.416	0.368	0.485	0.582	0.405

Three results follow. First, the raw LST association of Manavgat is above 0.5, and it falls well below 0.5 once elevation is held. This holds both for deciles and for removing the linear dependence of LST on elevation in the region (-5.6 K per km). On the full study area the same holds: 0.665 raw, against 0.455 and 0.446. The positive signal in Manavgat therefore comes from terrain, and with terrain held, Manavgat has the same sign as the other four regions. Second, in the other four regions the sign stays below 0.5 within elevation deciles in three regions and within NDVI deciles in all four. In Bejís it moves to 0.509 within elevation deciles and stays below 0.5 after detrending. Third, NDVI reverses in two regions once LST is held, but LST reverses in none of the four once NDVI is held. Aspect and illumination are not used, so terrain is held only through elevation.

S1.12 The LST anomaly under the difference instrument

Section 4.4 reports two elevation reversals on the collar, both involving Manavgat, that meet the per-comparison criterion. The LST anomaly differs between regions under a weaker test: the two regions are bootstrapped separately and their difference is taken. The three pairs below have point estimates on opposite sides of 0.5 and a difference interval that excludes zero, all for 1st_anomaly_mean. Signed AUC within the 10 km collar; the two regions are bootstrapped separately with 10-cell spatial blocks, 1000 replicates, seed 42.

Table S7. Between-region differences in the signed LST-anomaly association, 10 km collar. Pairs with point estimates on opposite sides of 0.5 and a difference interval that

excludes zero. Source `paper/labelfix_rerun/inference/reversal_family_holm.csv`
(collar frame); checked row by row by `paper/code/appendix_tables.py`.

Pair	AUC A	AUC B	Difference	95 % CI
Bejís vs Evia	0.392	0.584	−0.191	[−0.295, −0.083]
Evia vs Montiferru	0.584	0.400	+0.184	[+0.012, +0.321]
Bejís vs Muğla	0.392	0.507	−0.115	[−0.219, −0.002]

On the collar, 27 of the 90 feature-by-pair differences have intervals that exclude zero, against about 4.5 expected by chance at 5 %. Four of the 27 are for this feature. Three of these four have point estimates on opposite sides of 0.5, and two of the three involve Evia. With a Holm correction on normal-approximation p-values, 12 of the 90 remain. With the stricter intersection-union form, which also requires the association in each region to be significant, none remains (Section S1.20). The nine features carry about two independent thermal signals (Section 3.4), so the 90 comparisons are not independent. This result is weaker than a reversal.

S1.13 The same-geography event pair

Section 4.4 gives the result and shows why it does not survive the frame test. The design and two differences from the twenty-direction matrix are given here. The protocol is in Section S3.3.

This comparison was designed to keep the place fixed and change only the fire. Muğla burned in 2021 and again eleven months later, on the same grid and with the same processing. The 2022 population is the 2021 population without the 2021 scar: 41,730 rows with 2,911 burned for 2021, against 38,790 rows with 331 burned for 2022. **The 2021 population has 70 blocks of 5 km with burned cells, and the 2022 population has 11.** Eleven is below the minimum of sixteen used in this design, so the 2022 intervals are indicative, like the 20-cell row of Table 1. Values for all features are in Table S12.

326 **On the original study area, elevation reverses with bootstrap support.** In 2021, higher
 327 ground burned more, at 0.611 [0.532, 0.690]. In 2022, lower ground burned more, at 0.296
 328 [0.230, 0.355]. The intervals do not overlap, and the difference is -0.317 [-0.414 , -0.220].

329 **The collar removes this reversal.** The 2022 fire is one compact scar inside the whole Muğla
 330 rectangle, so 93.2 % of its cells lie more than 10 km from any burned cell (median 43.6 km),
 331 against 55.3 % in 2021. This is the largest share of distant cells in the cohort. On the 10 km
 332 collar, the 2021 value hardly changes (0.611 to 0.606, still supported). The 2022 value moves
 333 from 0.297 to 0.565, to the same side of 0.5 as in 2021, with an interval that includes chance.
 334 Slope does not reverse on either frame.

335 **Only elevation and slope can be evaluated.** The modelling export of the 2022 fire is not in
 336 this repository. The released script (paper/code/verify_mugla_collar.py) rebuilds the 2022
 337 population from the 2021 predictor file with the 2022 burned mask. This is valid for elevation,
 338 slope and land cover, which are the same in both years, but not for the seasonal channels.
 339 Compared with the pipeline's own 2022 export, the rebuilt data agree on elevation (0.297
 340 against 0.296) and slope (0.559 against 0.558). They differ on every seasonal channel: by
 341 $+0.136$ on `lst_anomaly_mean`, $+0.092$ on `tvdi_difference_mean`, -0.078 on `ndvi_mean` and
 342 -0.025 on `current_tvdi_mean`. **Only elevation and slope are therefore given a collar**
 343 **result here.**

Feature	2021, full	2022, full	full verdict	2021, collar	2022, collar	collar verdict
elevation_mean	0.611 [0.529, 0.692]	0.297 [0.229, 0.363]	supported reversal	0.606 [0.525, 0.685]	0.565 [0.450, 0.677]	none
slope_mean	0.637 [0.584, 0.689]	0.559 [0.463, 0.643]	none	0.635 [0.578, 0.692]	0.457 [0.356, 0.548]	none
the state- dependent channels	n/a	n/a	n/a	n/a	n/a	not evaluable here

Source paper/labelfix_rerun/code/mugla_two_event_collar.csv. The two populations are not separate samples. They share 38,789 of the 38,790 cells of the 2022 population. Elevation, slope and land cover are identical in all 73,098 grid cells, so only NDVI and the six thermal channels differ between them. The 2021 burned cells were also removed from the 2022 population. In the 2022-to-2021 direction, therefore, no burned target cell is in the source training data, while 38,789 of the 38,819 unburned target cells are. Membership of the training data alone separates the two classes of the 2021 target at ROC-AUC 0.9996. **The effect of this on a transfer estimate between the two years cannot be bounded.** The known part of the bias makes the reversal smaller, not larger: the removed cells are high (median 563 m), so removing them takes high unburned cells out of the 2022 population.

S1.14 The two interventions, in full

(a) Pooled multi-region training (Fig. 6). A model trained on the pooled populations of the other four regions is better than the mean of the four single-source models for one target of five, Evia, at 0.715 [0.668, 0.757] against a mean of 0.569. For the other four targets it is worse than this mean, by 0.009 to 0.057. It is better than the best single source only for Evia (0.654). The best source can only be chosen with target labels, so it is a reference, not an option. For every target the pooled model stays 0.20 to 0.48 AUC below the within-region ceiling. For Manavgat and Bejís it is below chance, and only for Manavgat with interval support, at 0.426 [0.369, 0.486]. Outside Evia, pooling does not recover what single-source transfer loses (paper/labelfix_rerun/code/loro_all.json, paper/labelfix_rerun/round7/r7d_pooled_vs_single.csv).

(b) Removing the direction-reversing features. Under the original label, two predictors reversed between regions with bootstrap support on the original study areas: **elevation_mean** and **1st_anomaly_mean**. This analysis keeps these two and does not re-select under the

corrected label (Table S10). Section 4.4 shows that the study area decides much of the support, so this analysis measures the cost of removal under the original protocol. Without the two predictors, mean within-region AUC falls by **-0.076**, with support in every region (-0.035, -0.130, -0.073, -0.063 and -0.079 for Manavgat, Bejís, Muğla, Evia and Montiferru; every interval fully below zero). Mean transfer changes by **+0.014 [-0.028, +0.056]**, which includes zero (source paper/labelfix_rerun/round3/feature_drop_transfer.json, summarised by paper/labelfix_rerun/round7/r7e_supplement_tables.py).

Two points should be noted. First, most of the cost does not come from the thermal set. Removing elevation alone costs -0.056 (0.896 to 0.840). Removing the LST anomaly alone costs -0.013 (0.896 to 0.883). About three quarters of the cost therefore comes from removing a baseline terrain variable. Second, both predictors were chosen because they reverse, on the same data on which the costs are estimated, and no correction is applied.

Table S28. Feature-removal configurations.

Configuration	Mean within-region AUC	Mean transfer AUC
full	0.896	0.527
drop elevation_mean	0.840	0.533
drop lst_anomaly_mean	0.883	0.529
drop both	0.820	0.541

Fig. 7 shows the four configurations together. A local cost is measured, but no transfer gain. This is not an exchange, because neither transfer effect is established: the thermal contribution to transfer is +0.007 and removing the reversing predictors gives +0.014, and both intervals include zero.

S1.15 Signed associations on both frames

Section 4.4 gives the collar results. The values per region are given here. Values per feature on both frames are in Table S14.

Reversals that meet the per-comparison criterion. On the original study areas, Table S10 counts fourteen supported reversals between pairs, across seven features. All but the anomaly pair involve Manavgat. On the collar, elevation is below 0.5 in Manavgat, at 0.376 [0.300, 0.465], and above 0.5 in the other four regions. It is supported in two of them: Muğla at 0.606 [0.525, 0.685] and Evia at 0.648 [0.550, 0.740] (paper/labelfix_rerun/code/collar_frame_bootstrap.csv). These two pairs are the only supported reversals left on the collar, and neither survives the Holm intersection-union correction (Section S1.20). The LST anomaly differs between regions only under the weaker difference test (Section S1.12).

The common sign. On the collar, LST is below 0.5 in four of five regions: 0.405, 0.332, 0.286 and 0.376 in Bejís, Muğla, Evia and Montiferru. A hotter pre-fire surface is therefore associated with less burning there, and the same holds for TVDI. Manavgat is the exception in the raw association (0.522 for LST, 0.527 for TVDI), but it shows the same sign once elevation is held (Section S1.11). Interval support differs: LST is supported in three of the four regions and TVDI in two, so “four of five agree” refers to point estimates. The absolute thermal channels therefore behave here as land-surface descriptors and not as a dryness index. The two differenced channels have no consistent direction across regions.

The number of independent reversals. The channels whose reversals disappear on the collar partly measure terrain. Across the regions, current LST is correlated with elevation at -0.695 to -0.125 , and TVDI at -0.722 to -0.298 . Within the collar, fused_1st_mean is correlated with current_1st_mean at 0.99 to 1.00, downscaled_1st_mean at 0.97 to 0.99 and current_tvdi_mean at 0.87 to 0.98. The two differenced channels are correlated at 0.64 to 0.94. Counts over the nine features therefore count features, not independent quantities. The thermal set carries about two independent signals.

The paired thermal contribution to transfer. The thermal and baseline matrices were differenced direction by direction. The mean is **+0.007**. The single contributions range from **−0.148 to +0.132**, with **twelve positive and eight negative**. A mean near zero therefore comes from positive and negative values that cancel, not from a consistent absence of effect. The directions are not independent, because each region appears in eight of the twenty. The interval therefore depends on the resampling unit. All four units that the design allows give intervals that include zero (1000 replicates where resampled; source `paper/labelfix_rerun/code/transfer_delta_ci.json`):

Table S29. The mean paired thermal contribution to transfer, by resampling unit.

Resampling unit	n	95 % interval on the mean paired contribution
Directions, naive bootstrap	20	[−0.023, +0.037]
Unordered pairs, cluster bootstrap (primary)	10	[−0.020, +0.038]
Unordered pairs, Student <i>t</i> on pair means	10	[−0.028, +0.043]
Regions, leave-one-out jackknife	5	[−0.018, +0.033]

Leaving out Manavgat, Bejís, Muğla, Evia and Montiferru in turn gives +0.0148, +0.0050, +0.0072, +0.0010 and +0.0087. No single region therefore carries the mean or changes its sign. These units resample only between directions. They do not include the sampling variability within a direction.

Precision. Across the twenty directions, the mean PR-AUC of the thermal model is **0.181**, **against a no-skill baseline of 0.157**. The mean of the twenty lifts over the baseline is 1.13. Seven of the twenty directions are below their own no-skill baseline at the point estimate. At 1 km blocking all seven have intervals fully below it. At 5 km blocking two do: Bejís to Manavgat and Muğla to Manavgat (`paper/labelfix_rerun/round7/r7a_pr_auc_10cell.csv`). Only one direction, Evia to Manavgat, is above twice its baseline, at 0.321 against 0.143 (Table S11).

The within-region gain on the collar. The baseline model was also run on the collar (paper/labelfix_rerun/round6/cosine/official_collar_increment_and_cosine.csv). The thermal gain at 5 km blocking is +0.073, +0.030, +0.088, +0.134 and +0.090 in the five regions. It is positive in all five, with a mean of +0.083, against +0.087 on the original study areas. On a 5 km collar the mean falls to +0.042, but it is still positive in all five regions. These are point estimates. The intervals of Table 1 refer to the original study areas.

The five study areas are not comparable. The share of modelled cells more than 10 km from any burned cell is 58.5 % in Manavgat, 63.1 % in Bejís, 55.3 % in Muğla, 43.7 % in Evia and 2.1 % in Montiferru. The median distances are 13.1, 13.5, 11.3, 8.0 and 2.7 km (Table S13). In Manavgat, the median elevation of modelled cells rises from 330 m within 5 km of the fire to 995 m at 10 to 20 km and 1,273 m at 20 to 50 km. The median elevation of the burned cells is 287 m.

S1.16 The contrast pair, in full

Manavgat and Muğla are in the same country, burned in the same year, and are 306 km apart. Their burned cells have the most similar environmental range of any pair in the matrix. Bejís and Montiferru have the least similar range (Fig. 8). Values for both pairs, on both frames, are in Table S18.

For Manavgat and Muğla on the original study areas, seven of nine feature-response directions point opposite ways, and all six features supported in both regions have opposite signs, including elevation. On the collar the elevation reversal remains, at 0.376 [0.300, 0.465] in Manavgat against 0.606 [0.525, 0.685] in Muğla. Transfer is below chance in both directions on both frames: 0.438 and 0.345 as drawn, 0.493 and 0.433 under the collar. On the collar, the interval of Muğla to Manavgat, [0.381, 0.491], excludes chance. The most similar pair is among the weakest in the matrix, but it is not the weakest. On the collar, the weakest

direction is Manavgat to Bejís at 0.407 and the strongest is Muğla to Evia at 0.728. The niche-overlap and applicability measures of Table S18 are computed on the original study areas.

Bejís and Montiferru are at the other extreme. Their burned ranges hardly overlap, and they have the most different values on every overlap measure. They transfer above chance in both directions, but only at the point estimate: neither direction has interval support at 5 km blocking. The applicability measure was not produced for Montiferru. **At the point estimates, high niche overlap did not ensure transfer, and low overlap did not prevent it.** This rests on two pairs. It is not a correlation across pairs; that is tested in Section S1.22.

S1.17 The sensitivity arms, summarised

Eight design choices were changed, one at a time, with everything else kept fixed: the Evia study area and its prevalence, the CORAL regularisation, the block size, the closing date of the predictor window, the quality screening of the coarse thermal input, normalised against absolute dryness channels, the removal of the coordinate-informed channels, and the capacity of the classifier (Sections S1.1 to S1.8). None changes a conclusion of the main text. Two of them affect how the results should be read.

The secondary population. Section 3.5 defines a secondary population of all valid cells, including cropland. It exists only for the within-region models in Manavgat and Bejís, so it is a two-region sensitivity analysis, and no transfer quantity is defined on it. Across the two regions and three block sizes, the thermal gain is +0.048 to +0.061, with every bootstrap interval above zero, against +0.045 to +0.067 in the primary population. The paired difference between the populations is −0.011 to +0.011, with no consistent sign (paper/labelfix_rerun/pipeline/robustness/step8_large_block_primary_all_valid/; Table 1). The gain is therefore not caused by excluding cropland. This analysis says nothing about transfer.

The block size. Larger blocks (5 km instead of 1 km) change the paired thermal-minus-baseline verdicts from twelve positive, seven negative and one uncertain to six, five and nine. The verdicts for the thermal model itself change from eleven above, seven below and two uncertain to nine, six and five (Section S1.3). Support is removed, never added. The point estimates do not change.

S1.18 Distance within a region

This analysis measures how skill falls with distance inside one region. A model is fitted on one half of a region and applied to the other half. Target cells are grouped by their distance from the training cells (paper/code/distance_curve.py; corrected output paper/labelfix_rerun/round3/distance_curve.json, summarised in paper/labelfix_rerun/round7/r7h_distance_curve.csv). Means are unweighted over bins. The bins contain 1 to 1,067 burned cells and have no intervals.

Table S32. Within-region skill by separation from the training cells.

Separation from training cells	Bins	Mean target AUC
0 to 5 km	18	0.709
5 to 10 km	16	0.570
10 to 20 km	11	0.519
20 to 40 km	6	0.499
40 to 80 km	3	0.541
80 to 160 km	1	0.421
cross-region, 306 to 2,802 km	20	0.527

At 10 to 20 km inside one region, the model is close to chance. This limits how far such a model can be used from its training cells. It agrees with Section 4.3, where withholding a scar or using a foreign model had no measurable cost.

This does not show that the cross-region failure is only an effect of distance. Once the curve reaches chance, any cross-region mean near 0.5 must lie on its continuation, so this comparison cannot test it. Also, a model that carries no information does not give a reliably

reversed ranking, but Manavgat to Bejís is below chance with interval support both on the original study area and on the 10 km collar.

Four limits apply. First, the two distance ranges do not overlap: within-region distances are 2 to 86 km, and cross-region distances start at 306 km. Second, the far bins contain few cells, so their means should not be read closely; the rise at 40 to 80 km is not evidence of an effect. Third, the near bins include the spatial autocorrelation that blocked validation is designed to remove, so **0.709 is an upper bound on near-field skill, not an estimate of it**. Fourth, the design separates distance from crossing a study-area boundary, but not from the land cover, terrain and fire history that change with distance. The analysis gives a length scale for the fall of skill within a region, not a cause.

S1.20 The frame test, elaborated

Section 4.4 gives these results. Sources: `paper/labelfix_rerun/code/aoi_frame_auc.csv` (signed associations by frame), `paper/labelfix_rerun/code/collar_frame_bootstrap.csv` (their 10-cell intervals), `paper/labelfix_rerun/round5/collar/aoi_frame_transfer.csv` (the transfer matrix by frame) and `paper/labelfix_rerun/code/matched_frame_gap.csv` (the matched reference); code in `paper/code/`.

Construction. Every region is restricted to cells within 10 km of any burned cell. This removes only distant unburned cells. Every burned cell is at distance zero and is kept at any radius, so radii were compared to guard against a favourable choice. The interval criterion was evaluated at the 10 km collar; the 5 and 10 km radii agree at the point estimate. The collar strongly reduces the number of cells in two regions, so part of any loss of support is a loss of power. On average, 2.3 features per direction are supported in both regions on the original study areas, and 2.2 on the collar.

Multiplicity. Each frame has a family of ninety feature-by-pair comparisons (nine features, ten region pairs). Table S30 gives how many pass each criterion. The per-comparison criterion of Section 3.10 leaves two supported reversals on the collar, both on elevation and both involving Manavgat. Neither survives the Holm intersection-union correction, which adjusts the tests of both regions together (adjusted $p = 0.33$ for Manavgat against Evia and 0.79 for Manavgat against Muğla, normal approximation).

Table S30. Reversal family, ninety feature-by-pair comparisons per frame. Source `paper/labelfix_rerun/inference/reversal_family_holm.csv`, `reversal_per_region.csv`.

Criterion	Full frame	10 km collar
Difference interval excludes zero	39	27
... with opposite-sided point estimates	31	21
Per-comparison criterion (each region's own interval excludes 0.5, opposite sides)	14	2
Holm, normal approximation, difference test	27	12
Holm intersection-union (both regions' tests)	5	0

Similarity measures on the collar. Two of the twenty similarity measures are built from the signed associations, and they were recomputed on the collar. The sign-agreement fraction over supported features is $\rho = +0.52 [-0.27, +0.87]$ on the original study areas. On the collar it takes only the values 0 and 1 across sixteen defined directions, and it correlates with collar transfer at $\rho = +0.38 [-0.17, +0.85]$. The all-feature cosine is $+0.44 [-0.29, +0.80]$ on the original study areas and correlates with collar transfer at **$+0.57 [+0.05, +0.88]$** . This is the only interval of a similarity measure in the study that excludes zero (paper/labelfix_rerun/round6/diag_collar/; intervals from paper/labelfix_rerun/round7/r7g_collar_diagnostics.json, which reproduces the original intervals exactly). It is not treated as a result for three reasons. It is one of about forty uncorrected tests. It rests on twenty directions that share regions. The collar is defined from

543 the burned cells, so the measure needs target labels twice. The other eighteen measures were
544 not recomputed on the collar.

545 **Data provenance.** A quality-screening rebuild overwrote the 500 m modelling datasets of
546 Manavgat and Muğla at the standard pipeline path after the frozen tables were computed.
547 Each new file differs in `downscaled_1st_mean` and `fused_1st_mean`. The pipeline records a
548 SHA-256 for the modelling dataset of each region, and this hash identifies the frozen copy.
549 Every analysis of Section 4.4 reads each region through `paper/code/_canonical.py`, which
550 checks the hash on load, and reads Manavgat from the corrected re-freeze (SHA-256
551 `5a5e876c...`). The provenance of the collar transfer matrix is in
552 `paper/labelfix_rerun/round5/collar/PROVENANCE.md`.

553 **Table S8. Cross-region transfer on comparable study areas.** Natural-vegetation
554 population, thermal model, twenty ordered directions per row. Above and below chance are
555 point counts. The supported counts use a 10-cell (≈ 5 km) spatial-block bootstrap on the target,
556 1000 replicates, seed 42. Table S16 gives the original matrix under 2-cell (≈ 1 km) blocking.
557 Bounds per direction are in
558 `paper/labelfix_rerun/round5/collar/aoi_frame_transfer.csv`.

Source frame	Target frame	Mean target AUC	Above chance	Below chance	Supported above / below	Paired thermal delta
full	full	0.527	13 of 20	7	9 / 6	+0.007
full	10 km	0.559	14 of 20	6	10 / 2	+0.008
10 km	full	0.546	14 of 20	6	10 / 5	+0.014
10 km	10 km	0.589	15 of 20	5	13 / 2	+0.024
5 km	5 km	0.591	16 of 20	4	11 / 0	+0.017

559 The baseline changes with the frame. On the 10 km collar, the static baseline transfers at
560 0.565 and the thermal model at 0.589, a paired difference of +0.024 instead of +0.007. The
561 static baseline still does not transfer well. The number of directions below chance falls from
562 seven to five at the point estimate, and from six to two with interval support. The largest rises

are Bejís to Evia (0.383 to 0.602) and Bejís to Manavgat (0.314 to 0.452). The only large fall is Evia to Manavgat (0.677 to 0.594).

The matched reference. The within-region reference was recomputed on the same frame at 5 km blocking (paper/code/verify_matched_gap.py):

Table S33. Within-region reference and transfer on matched frames.

Frame	Within-region (5 km blocking)	Mean transfer	Gap
full rectangle	0.814	0.527	0.287
10 km collar	0.786	0.589	0.197
5 km collar	0.759	0.591	0.169

Paired by target region, the shortfall on the collar is **+0.197 [+0.091, +0.303]** (Student *t* over the five target regions; +0.081 Montiferru, +0.319 Manavgat, +0.172 Muğla, +0.202 Bejís, +0.211 Evia). The shortfall remains on every matched row. It is 0.197 on the collar, not the 0.29 of the unmatched comparison, and it becomes smaller as the frame moves closer to the fire. On the 5 km collar, the within-region gain falls by half but stays positive in all five regions, with a mean of +0.042 (Manavgat +0.035).

S1.21 The transfer matrix and adaptation, elaborated

Section 4.5 gives these results.

Raw transfer. Raw target AUC ranges from 0.314 to 0.677 (Fig. 4). At 2-cell blocking, eleven of 20 directions are above chance with interval support and seven are below: both directions between Manavgat and Bejís, both between Manavgat and Muğla, both between Bejís and Evia, and Montiferru to Manavgat. At 10-cell blocking the same point estimates give 9 above, 6 below and 5 uncertain. Manavgat to Muğla loses its support, and no direction changes side of the chance line. Of the directions below chance, Manavgat to Bejís (0.407 [0.323, 0.493]) and Muğla to Manavgat (0.433 [0.381, 0.491]) keep interval support on the 10

km collar, but both intervals include chance on the 5 km collar. The lowest direction on the original study areas is Bejís to Manavgat, at 0.314 [0.296, 0.332].

Label-blind adaptation. Under region-wise z-scoring the twenty directions range from 0.302 to 0.630, and under CORAL from 0.406 to 0.624. The change is not uniform: under z-scoring, Bejís to Manavgat falls further below chance, from 0.314 to 0.302. With the better of the two methods in each direction, 15 of the 20 directions end closer to chance and 5 end further from it. Four of these five involve Montiferru and move up. The fifth, Manavgat to Muğla, moves down from 0.438 to 0.427. Under CORAL alone, sixteen of twenty end closer to chance (paper/labelfix_rerun/round5/matrix20_official.csv). In the six directions where raw transfer was below chance, the best label-free method recovers at most 28 % of the gap to the within-region reference (Bejís to Evia). On the twelve directions of the decomposition, seven show negative recovery (Section S1.10).

Seed stability. Across five bootstrap seeds, every 2-cell verdict is stable. At 10 cells, one level verdict and two paired-delta verdicts change: the level verdict of Evia to Bejís, and the paired-delta verdicts of Manavgat to Muğla and Evia to Montiferru (paper/labelfix_rerun/round6/seed_stability.json). The random-forest seed is fixed, so all intervals per direction are conditional on one fitted source model.

Jackknife. Leaving out one region at a time moves the mean paired contribution between +0.001 (without Evia) and +0.015 (without Manavgat). Its sign never changes.

S1.22 The similarity diagnostics

Twenty similarity measures from five families were each rank-correlated with raw thermal transfer AUC over the directions on which they are defined. One pair-based bootstrap is used for all of them (unordered pairs resampled with replacement, both directions kept, 2000 replicates, seed 42). The measures were defined on 8 August 2026, together with their results

607 under the original label, and were not re-selected after the label correction. Table S17 lists all
608 twenty. The source is
609 paper/labelfix_rerun/round5/out_official/all_diagnostics_vs_transfer.csv, and
610 the original-label values are in paper/labelfix_rerun/round5/s6_diagnostics_20.md.

611 **On the original study areas, no interpretable measure has an interval that excludes zero.**

612 The largest correlations are for the conditional family: the sign-agreement fraction over
613 supported features, at $\rho = +0.52$ $[-0.27, +0.87]$, and its cosine, at $+0.49$ $[-0.24, +0.87]$. Under
614 the original label these two were the only rows with intervals that excluded zero ($+0.84$ and
615 $+0.81$). This did not hold after the label correction. The marginal family (applicability,
616 dissimilarity, climatic distance, domain classification), the niche-overlap family and the
617 regime family all include zero. Geographic distance does not order the matrix ($\rho = -0.17$
618 $[-0.87, +0.84]$ on the twelve directions where it is defined). The domain classifier separates
619 source from target at $AUC \geq 0.96$ for every pair. Because it always succeeds, it gives no
620 ordering. The twentieth measure, the vector Spearman of signed AUCs over supported
621 features, is defined on only six directions. Its interval is degenerate (the upper bound equals
622 the point estimate), so it is listed but not interpreted.

623 **Equal samples.** The families use different samples: twelve directions for the marginal,
624 applicability, climatic and geographic rows, eighteen for the supported conditional rows, and
625 twenty for the others. Every row was therefore recomputed on the common subsets of twelve
626 and sixteen directions. On both, every interpretable row includes zero
627 (paper/labelfix_rerun/round5/out_official/diagnostics_common_subset.json).

628 **The collar.** Section S1.20 gives the two measures recomputed on the collar. One of them, the
629 all-feature cosine, orders collar transfer with an interval that excludes zero. It is reported with
630 its limits and is not used.

Limits. Signed associations need burned labels in both regions, so the conditional family cannot be used before a fire. The marginal family can be used, but it fails. Ten independent region pairs give little power, and a moderate ordering would usually be missed. These results therefore show a failure to find an ordering, not proof that none exists (Section S3.5(xiii)). Measures based on interval support are unstable when bounds lie near 0.5 (Section S3.5(viii)).

S1.23 Additional robustness analyses

The analyses in Table S21 were run after the main results. They test whether the main quantities depend on analysis choices. None of them was used to select a reported configuration.

Table S21. Additional robustness analyses (post hoc). Sources under paper/labelfix_rerun/: geometry/ (a1 to a5), inference/ (units, equivalence, ladder) and round7/.

Analysis	Question	Result
Edge-excluded frame cost (a1)	Is the frame cost carried by the scar's edge cells?	Over nine scars, excluding 0, 1 or 2 cells of the scar edge gives $A - B = 0.147$ [0.086, 0.208], 0.137 [0.069, 0.205] and 0.129 [0.055, 0.204]; with the region as unit 0.160, 0.150 and 0.145, every interval above zero
Placebo collars (a2)	Is the cost a property of any fire-shaped frame?	The scar's real positives scored against the negatives of a same-shaped collar placed away from any burned cell reproduce A ($A - \text{placebo} -0.002$ [-0.056, +0.052], eight scars with placements); the real collar costs +0.155 [+0.108, +0.202] more, so the cost is the fire-adjacent negatives
Label-free target frames (a4)	Does a frame drawn without target labels change transfer?	Trimming the target to the source's per-feature range, or to the source's area of applicability (Meyer and Pebesma 2021), gives means of 0.520 to 0.574, and a 20 km window 0.482; every paired thermal Δ interval spans zero
Metric dependence (a5)	Is the frame cost specific to ROC-AUC?	Region-wide minus scar-frame over nine scars: partial AUC ($\text{FPR} \leq 0.1$) 0.088 [0.048, 0.128]; average precision -0.342 [-0.477, -0.206], higher on the scar frame because its prevalence is higher
Collar within-region increment	Does the within-region increment survive the collar?	+0.083 [+0.037, +0.129] over five regions (Student t), against +0.087 [+0.037, +0.136] as drawn

Analysis	Question	Result
Equalised Δ by resampling unit	Does the collar Δ exclude zero?	Table S31: two of five computable units exclude zero; the two-way estimators are undefined
Equivalence of the equalised Δ	How large a gain is excluded?	Within ± 0.05 under four of five units; no gain above 0.047 at 90 % (pair cluster); not within ± 0.02 under any unit
Within minus transfer increment	Is the local-versus-portable contrast itself supported?	Full frame +0.079 [+0.014, +0.145]; 10 km collar +0.059 [+0.007, +0.110]; 5 km collar +0.026 [−0.012, +0.063]
Reversal multiplicity	Do the collar reversals survive correction?	Table S30: two per-comparison reversals, none after Holm intersection-union
scikit-learn version	Do point estimates depend on the library version?	Under 1.5.2 against 1.9.0, single directions move by up to 0.047 and means by at most 0.003; one support count changes (round7/sklearn152/compare_vs_1_9_0.json)

643 **Table S31. The equalised (10 km collar) paired contribution by resampling unit.** 20,000
644 replicates where resampled; 95 % intervals. Source
645 paper/labelfix_rerun/inference/units.json.

Unit	95 % interval	Excludes zero
Directions, naive	[−0.001, +0.049]	no
Unordered pairs, cluster bootstrap (primary)	[−0.003, +0.051]	no
Target region, cluster bootstrap	[+0.010, +0.040]	yes
Source region, cluster bootstrap	[+0.010, +0.042]	yes
Pigeonhole (source and target)	[−0.005, +0.060]	no
Two-way cluster (CGM)	undefined (negative variance)	n/a
Dyadic-robust	undefined (negative variance)	n/a
Leave-one-region-out jackknife, t	[−0.017, +0.065]	no

646 The primary unit is the pair cluster (Section 3.7), and on it the paired contribution on the
647 collar includes zero. With five regions, a percentile bootstrap over regions has at most 126
648 distinct resamples, so the two region-clustered intervals are coarse.

649 S2 Supporting tables

650 These tables give the values per region and per direction that the Results quote, so that every
651 statement can be checked against the values behind it.

Table S9. Signed univariate AUC of each predictor against burned, by region. Natural-vegetation population; 10-cell (~5 km) spatial-block bootstrap, 1000 replicates, seed 42. The AUC is not folded to $\max(\text{AUC}, 1 - \text{AUC})$. A value below 0.5 means that lower values rank burned cells higher; it shows a direction, not weakness. **Bold** marks a region whose own interval excludes 0.5. Source: `paper/labelfix_rerun/round5/tables/step9g_multi_aoi_feature_stability.csv` (the pipeline's Step9G five-region synthesis, sha256 c864cd7d...); checked row by row by `paper/code/appendix_tables.py`.

Feature	Manavgat	Bejís	Muğla	Evia	Montiferru
elevation_mean	0.232 [0.179, 0.288]	0.643 [0.558, 0.729]	0.611 [0.532, 0.690]	0.541 [0.448, 0.626]	0.584 [0.395, 0.762]
slope_mean	0.400 [0.340, 0.459]	0.521 [0.439, 0.605]	0.637 [0.582, 0.686]	0.487 [0.418, 0.554]	0.652 [0.506, 0.771]
ndvi_mean	0.564 [0.499, 0.628]	0.559 [0.497, 0.619]	0.662 [0.616, 0.704]	0.639 [0.575, 0.701]	0.586 [0.450, 0.704]
lst_anomaly_mean	0.509 [0.460, 0.560]	0.418 [0.364, 0.480]	0.485 [0.395, 0.566]	0.640 [0.567, 0.710]	0.395 [0.285, 0.535]
current_lst_mean	0.665 [0.608, 0.719]	0.477 [0.401, 0.547]	0.325 [0.271, 0.382]	0.377 [0.301, 0.456]	0.370 [0.248, 0.513]
current_tvdi_mean	0.677 [0.622, 0.733]	0.517 [0.429, 0.595]	0.336 [0.275, 0.398]	0.362 [0.285, 0.442]	0.356 [0.233, 0.499]
tvdi_difference_mean	0.460 [0.409, 0.510]	0.512 [0.443, 0.583]	0.490 [0.396, 0.575]	0.519 [0.444, 0.589]	0.378 [0.282, 0.497]
downscaled_lst_mean	0.683 [0.626, 0.739]	0.484 [0.400, 0.560]	0.307 [0.253, 0.366]	0.377 [0.297, 0.459]	0.365 [0.240, 0.511]
fused_lst_mean	0.666 [0.610, 0.721]	0.481 [0.404, 0.551]	0.325 [0.272, 0.383]	0.376 [0.300, 0.456]	0.370 [0.248, 0.513]

Definition of a reversal. A pair of regions is called a reversal only when their signed associations lie on opposite sides of 0.5 **and the interval of each region excludes 0.5**. This is stricter than requiring disjoint intervals. For example, for `current_lst_mean` in Manavgat and Bejís the two intervals are disjoint ([0.608, 0.719] against [0.401, 0.547]), but the Bejís interval includes 0.5. Bejís therefore has no established direction, and this pair is a point reversal only.

Table S10. Cross-region reversals that meet the strict criterion. The criterion is applied to the intervals of Table S9. Difference intervals are from the paired 10-cell block bootstrap of `paper/code/ems_inference_multiplicity.py` (`paper/labelfix_rerun/inference/reversal_family_holm.csv`, 1000 replicates), which flags the same fourteen pairs. Checked row by row by `paper/code/appendix_tables.py`.

Feature	Region A	AUC	Region B	AUC	Difference [95 % CI]
elevation_mean	Manavgat	0.232	Bejís	0.643	+0.411 [+0.312, +0.509]
elevation_mean	Manavgat	0.232	Muğla	0.611	+0.379 [+0.274, +0.476]
slope_mean	Manavgat	0.400	Muğla	0.637	+0.237 [+0.158, +0.313]
slope_mean	Manavgat	0.400	Montiferru	0.652	+0.252 [+0.100, +0.389]
lst_anomaly_mean	Bejís	0.418	Evia	0.640	+0.222 [+0.129, +0.316]
current_lst_mean	Muğla	0.325	Manavgat	0.665	+0.340 [+0.260, +0.413]
current_lst_mean	Evia	0.377	Manavgat	0.665	+0.288 [+0.195, +0.384]
current_tvdi_mean	Muğla	0.336	Manavgat	0.677	+0.341 [+0.253, +0.422]
current_tvdi_mean	Evia	0.362	Manavgat	0.677	+0.315 [+0.221, +0.413]
current_tvdi_mean	Montiferru	0.356	Manavgat	0.677	+0.322 [+0.161, +0.453]
downscaled_lst_mean	Muğla	0.307	Manavgat	0.683	+0.376 [+0.295, +0.451]
downscaled_lst_mean	Evia	0.377	Manavgat	0.683	+0.307 [+0.209, +0.405]
fused_lst_mean	Muğla	0.325	Manavgat	0.666	+0.341 [+0.261, +0.413]
fused_lst_mean	Evia	0.376	Manavgat	0.666	+0.291 [+0.198, +0.386]

There are fourteen reversals between pairs, across **seven** features, and thirteen of them involve Manavgat. Under the original label there were three, across two features: elevation and the LST anomaly. The feature-removal analysis of Section 3.11 keeps these two and does not re-select. Twenty-six further pairs reverse at the point estimate only, and they are not counted. The strict criterion reduces the number of findings: a difference interval on the pair (the test of Section S1.12) would support seventeen further reversals. None of the fourteen is corrected for multiplicity; Table S30 gives the corrected counts.

S2.1 The transfer matrix in precision-recall terms

ROC-AUC is used in the main text so that the results can be compared with the susceptibility literature. A susceptibility map is used to rank areas, so precision-recall measures its practical value. At target prevalences of 7.0 to 28.7 %, the two measures can differ strongly. Values are

682 read from the step9b exports of the re-frozen outputs
683 (paper/labelfix_rerun/round5/tables/corrected/*/step9b_metrics.json).

684 **Table S11. Thermal transfer, PR-AUC against the no-skill baseline.** The baseline is the
685 burned prevalence of the target. Lift is PR-AUC divided by this baseline; a lift of 1 is no
686 better than a random ranking. Ordered by lift. Checked row by row by
687 paper/code/appendix_tables.py. PR-AUC intervals at 2-cell and 10-cell blocking for every
688 direction are in paper/labelfix_rerun/round7/r7a_pr_auc_10cell.csv.

Direction	ROC-AUC	PR-AUC	No-skill	Lift
Evia → Manavgat	0.677	0.321	0.143	2.24
Manavgat → Evia	0.654	0.407	0.287	1.42
Bejís → Montiferru	0.594	0.289	0.212	1.36
Evia → Montiferru	0.647	0.283	0.212	1.34
Bejís → Muğla	0.618	0.093	0.070	1.33
Muğla → Evia	0.653	0.379	0.287	1.32
Montiferru → Bejís	0.548	0.093	0.072	1.28
Montiferru → Muğla	0.619	0.089	0.070	1.27
Evia → Muğla	0.577	0.086	0.070	1.23
Muğla → Bejís	0.583	0.088	0.072	1.22
Manavgat → Montiferru	0.518	0.243	0.212	1.15
Montiferru → Evia	0.586	0.317	0.287	1.11
Muğla → Montiferru	0.531	0.214	0.212	1.01
Manavgat → Muğla	0.438	0.060	0.070	0.85
Evia → Bejís	0.448	0.059	0.072	0.82
Montiferru → Manavgat	0.404	0.113	0.143	0.79
Bejís → Evia	0.383	0.226	0.287	0.79
Manavgat → Bejís	0.396	0.054	0.072	0.75
Muğla → Manavgat	0.345	0.101	0.143	0.70
Bejís → Manavgat	0.314	0.098	0.143	0.68
Mean	0.527	0.181	0.157	1.13

689 Seven directions are below their own no-skill baseline, and only one is above twice its
690 baseline. The seven are the same seven that are below chance on ROC-AUC, as expected for
691 a reversed ranking. At 1 km blocking, all seven have PR-AUC intervals fully below the
692 baseline. At 5 km blocking two do: Bejís to Manavgat [0.067, 0.132] and Muğla to Manavgat
693 [0.071, 0.135], against 0.143. Section 4.4 shows that the number of directions below chance
694 depends largely on the study areas.

S2.2 The same-geography event pair, in full

Section S1.13 reports this comparison. Section 4.4 shows that its elevation reversal is caused by the study area, and Section S1.13 explains why its thermal channels are not evaluated. The values per feature are kept here because this comparison led to the frame test.

Table S12. Signed univariate feature-burned AUC, Muğla 2021 versus 2022. Raw AUC against burned, not folded to $\max(\text{AUC}, 1 - \text{AUC})$; 10-cell (≈ 5 km) spatial-block bootstrap, 1,000 replicates, seed 42 (Section 3.14). Analysis population 41,730 rows / 2,911 burned (2021) and 38,790 rows / 331 burned (2022). **Blocks of 5 km with burned cells: 70 in 2021 and 11 in 2022.** The note to Table 1 sets sixteen as the minimum for this blocking, so the 2022 intervals are indicative, like the 20-cell row of Table 1. The 2022 fire is also one compact scar, so its eleven blocks are next to each other. Read from `paper/step9g_raw/.../mugla_2021__mugla_2022_event_relative/step9g_direction_reversal_table.csv`.

Feature	2021 AUC [95 % CI]	2022 AUC [95 % CI]	Reversal
elevation_mean	0.611 [0.532, 0.690]	0.296 [0.230, 0.355]	bootstrap-supported
current_lst_mean	0.325 [0.271, 0.382]	0.515 [0.434, 0.580]	point only
current_tvdi_mean	0.336 [0.275, 0.398]	0.594 [0.475, 0.674]	point only
downscaled_lst_mean	0.307 [0.253, 0.366]	0.508 [0.435, 0.571]	point only
fused_lst_mean	0.325 [0.272, 0.383]	0.519 [0.436, 0.583]	point only
ndvi_mean	0.662 [0.616, 0.704]	0.707 [0.624, 0.777]	none
slope_mean	0.637 [0.582, 0.686]	0.558 [0.468, 0.634]	none
lst_anomaly_mean	0.485 [0.395, 0.566]	0.380 [0.249, 0.502]	none
tvdi_difference_mean	0.490 [0.396, 0.575]	0.397 [0.265, 0.514]	none

S2.3 The evaluation frames, and the signed associations they produce

Section 4.4 gives these results. The values per region are given here.

Table S13. Geometry of the five study areas. Natural-vegetation population. Distance is the Euclidean distance to the nearest burned cell on the 500 m grid, at 0.45 km per cell. Corrected Manavgat label. Computed by `paper/code/appendix_tables.py` from the step8a tables (read

713 through `paper/code/_canonical.py`, which checks the sha256 of each file), as in
 714 `paper/code/verify_aoi_frame.py`.

Region	cells	burned	median distance to burned	share beyond 10 km
Manavgat	20,511	2,935	13.1 km	58.5 %
Bejís	15,190	1,100	13.5 km	63.1 %
Muğla	41,730	2,911	11.3 km	55.3 %
Evia	9,298	2,664	8.0 km	43.7 %
Montiferru	2,544	539	2.7 km	2.1 %

715 **Table S14. Signed univariate AUC, original study area against a 10 km collar.** Point
 716 estimates. The intervals used for the reversal test are given in the text and in
 717 `collar_frame_bootstrap.csv` (10-cell blocks, 1000 replicates, seed 42). Signed and not
 718 folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 shows a direction, not weakness. The
 719 collar removes no burned cells in any region. Full frame from the source of Table S9; collar
 720 from `paper/labelfix_rerun/code/aoi_frame_auc_frozen_mugla.csv` (the collar analysis
 721 with Muğla read from its canonical file). Checked row by row by
 722 `paper/code/appendix_tables.py`. On every row, Manavgat is on the other side of 0.5 from
 723 the other regions, also on the collar.

Signed univariate AUC	Manavgat	Bejís	Muğla	Evia	Montiferru	straddles 0.5
elevation, full frame (Table S9)	0.232	0.643	0.611	0.541	0.584	yes
elevation, 10 km collar	0.376	0.614	0.606	0.648	0.581	yes
current LST, full frame	0.665	0.477	0.325	0.377	0.370	yes
current LST, 10 km collar	0.522	0.405	0.332	0.286	0.376	yes
current TVDI, full frame	0.677	0.517	0.336	0.362	0.356	yes
current TVDI, 10 km collar	0.527	0.454	0.342	0.250	0.361	yes

724 **Table S15. Region summary: populations and gate results.** Counts from the Step 8A
 725 dataset statistics of each region; gate fractions from the burned-landcover gate output of each
 726 region. TSG = the natural-vegetation population (tree, shrub and grass). The TSG columns
 727 use the modelled population, `burnable_tree_shrub_grass` **and** `valid_for_modeling ==`
 728 `True`, on which every model in this paper was fitted and scored. Counts are computed from
 729 the step8a tables, and gate fractions are read from the gate output of each region

730 (paper/labelfix_rerun/round5/tables/corrected/gates/). Checked row by row by
 731 paper/code/appendix_tables.py.

Region	Total cells	Valid cells	Burned	Prevalence (all valid)	TSG cells	Burned in TSG	TSG prevalence	Burned natural-veg fraction	Gate verdict
Manavgat 2021	24,150	24,087	3,046	0.126	20,511	2,935	0.143	0.955	pass
Bejís 2022	15,759	15,759	1,103	0.070	15,190	1,100	0.072	0.991	pass
Muğla 2021	73,098	73,045	3,026	0.041	41,730	2,911	0.070	0.958	pass
North Evia 2021 (extended)	22,925	22,906	2,788	0.122	9,298	2,664	0.287	0.945	pass
Montiferru 2021	3,234	3,173	697	0.220	2,544	539	0.212	0.723	pass

732 **Table S16. Cross-region transfer matrix, thermal model, TSG population.** Target ROC-
 733 AUC with 2-cell spatial-block bootstrap 95% CIs (1000 replicates). CORAL is applied after
 734 region-wise z-scoring ($\lambda = 10^{-5}$). Generated by paper/code/table_b9.py from the re-frozen
 735 outputs (paper/labelfix_rerun/round5/matrix20_official.csv).

Direction	Raw	Region-wise z-score	CORAL
Manavgat→Bejís	0.396 [0.373, 0.422]	0.450 [0.423, 0.477]	0.467 [0.441, 0.491]
Bejís→Manavgat	0.314 [0.296, 0.332]	0.302 [0.282, 0.323]	0.406 [0.388, 0.423]
Manavgat→Muğla	0.438 [0.418, 0.456]	0.427 [0.408, 0.444]	0.417 [0.398, 0.436]
Muğla→Manavgat	0.345 [0.331, 0.359]	0.485 [0.468, 0.502]	0.476 [0.460, 0.493]
Manavgat→Evia	0.654 [0.633, 0.676]	0.529 [0.504, 0.553]	0.504 [0.481, 0.528]
Evia→Manavgat	0.677 [0.658, 0.697]	0.404 [0.386, 0.421]	0.417 [0.399, 0.435]
Bejís→Muğla	0.618 [0.601, 0.635]	0.518 [0.501, 0.535]	0.507 [0.489, 0.524]
Muğla→Bejís	0.583 [0.561, 0.607]	0.535 [0.512, 0.557]	0.560 [0.538, 0.581]
Bejís→Evia	0.383 [0.363, 0.402]	0.532 [0.509, 0.551]	0.499 [0.479, 0.518]
Evia→Bejís	0.448 [0.426, 0.470]	0.549 [0.524, 0.575]	0.549 [0.525, 0.573]
Muğla→Evia	0.653 [0.636, 0.671]	0.561 [0.543, 0.580]	0.563 [0.545, 0.582]
Evia→Muğla	0.577 [0.560, 0.593]	0.501 [0.485, 0.518]	0.530 [0.515, 0.546]
Montiferru→Manavgat	0.404 [0.385, 0.422]	0.388 [0.367, 0.408]	0.436 [0.416, 0.456]
Manavgat→Montiferru	0.518 [0.474, 0.563]	0.527 [0.484, 0.568]	0.505 [0.461, 0.547]
Montiferru→Bejís	0.548 [0.521, 0.578]	0.574 [0.552, 0.596]	0.569 [0.548, 0.591]
Bejís→Montiferru	0.594 [0.560, 0.631]	0.550 [0.500, 0.601]	0.574 [0.530, 0.621]
Montiferru→Muğla	0.619 [0.604, 0.634]	0.576 [0.562, 0.589]	0.565 [0.550, 0.579]
Muğla→Montiferru	0.531 [0.495, 0.568]	0.587 [0.549, 0.624]	0.584 [0.547, 0.623]
Montiferru→Evia	0.586 [0.565, 0.606]	0.630 [0.611, 0.649]	0.624 [0.605, 0.641]
Evia→Montiferru	0.647 [0.608, 0.682]	0.568 [0.528, 0.609]	0.581 [0.539, 0.623]

736 S2.4 The transferability-diagnostics tables

737 Table S17. All similarity measures against raw thermal transfer (20 ordered directions).

738 Spearman ρ with pair-based bootstrap 95 % CIs. Exp. = expected sign. Rows with $n = 12$ exist
739 only for the four-region subset, because these measures were not produced for Montiferru.
740 The supported conditional rows use the 18 directions with at least one supported feature.
741 Every row is from `paper/labelfix_rerun/round5/s6_diagnostics_20.csv`, checked by
742 `paper/code/appendix_tables.py`. The twenty measures were defined on 8 August 2026 and
743 were not re-selected after the label correction.

Diagnostic	Family	Exp.	n dir	Spearman ρ [95% CI]	CI excludes 0
Agreement fraction, supported features	P(y x) conditional	+	18	+0.52 [-0.27, +0.87]	no
Cosine, supported features	P(y x) conditional	+	18	+0.49 [-0.24, +0.87]	no
Cosine, all 9 features	P(y x) conditional	+	20	+0.44 [-0.29, +0.80]	no
Vector Spearman, all 9	P(y x) conditional	+	20	+0.43 [-0.21, +0.80]	no
Agreement count, all 9	P(y x) conditional	+	20	+0.47 [-0.29, +0.80]	no
Schoener's D, 1-D mean	P(x y=1) niche	+	20	+0.10 [-0.53, +0.63]	no
Warren's I, 1-D mean	P(x y=1) niche	+	20	+0.19 [-0.43, +0.77]	no
Schoener's D, PCA-2D	P(x y=1) niche	+	20	+0.03 [-0.59, +0.63]	no
Warren's I, PCA-2D	P(x y=1) niche	+	20	-0.03 [-0.66, +0.56]	no
Mahalanobis, burned centroids	P(x y=1) niche	-	20	-0.22 [-0.82, +0.42]	no
Domain-classifier AUC	P(ix) marginal	-	20	-0.33 [-0.81, +0.36]	no
Predictor-space mean dissimilarity	P(ix) marginal	-	12	-0.13 [-0.67, +0.37]	no
Predictor-space p95 dissimilarity	P(ix) marginal	-	12	-0.15 [-0.68, +0.24]	no
Fraction inside weighted AoA	P(ix) marginal	+	12	+0.21 [-0.48, +0.62]	no
Fraction inside unweighted support	P(ix) marginal	+	12	+0.08 [-0.92, +0.76]	no
Climatic distance	P(ix) marginal	-	12	+0.03 [-0.84, +0.84]	no

Diagnostic	Family	Exp.	n dir	Spearman ρ [95% CI]	CI excludes 0
Geographic distance	geographic	–	12	–0.17 [–0.87, +0.84]	no
Regime distance, log effective- N	P(y) structure	–	20	+0.11 [–0.67, +0.80]	no
Regime distance, largest share	P(y) structure	–	20	+0.11 [–0.66, +0.77]	no
Vector Spearman, supported (≥ 3 feats) ‡	P(y x) conditional	+	6	+0.96 [+0.85, +0.96]	degenerate interval; not interpreted

Table note. A null row means that the measure was **not shown to order transfer** in this design. It does not mean that the measure cannot order transfer. With ten independent region pairs the power is low, and the intervals are wide enough to include moderate true correlations in either direction. ‡ The last row belongs to the set of measures fixed in advance, so it stays in the table. It is defined on six directions only, and its interval is degenerate (the upper bound equals the point estimate), so it is not interpreted and cannot be compared with the other nineteen.

Under the original label, the two supported-feature variants excluded zero (+0.84 and +0.81 over sixteen directions). Both choose their predictors by whether the bootstrap intervals of two regions exclude 0.5, a choice made on the same data. Under the corrected label neither excludes zero (+0.52 and +0.49 over eighteen directions), and no interpretable row does (see Section S3.5(viii) on why measures based on interval support are unstable).

Equal-sample check. The families use different samples: twelve directions for the marginal, applicability, climatic and geographic rows, eighteen for the supported conditional rows, and twenty for the others. Every row was recomputed on the common subset of twelve directions, and again on the sixteen directions where the supported conditional rows are defined. Every interpretable row still includes zero. On the common twelve, the supported conditional rows give +0.41 [–0.42, +0.88] and +0.46 [–0.56, +0.84]. The published values are reproduced to 4.4×10^{-5} . Source:

paper/labelfix_rerun/round5/out_official/diagnostics_common_subset.json.

Table S18. The most and least environmentally similar pairs, on both frames. Schoener's D is computed over burned cells only, so it does not change with the collar. Transfer values are the two ordered directions of each pair. Ranks are out of the twenty directions on the collar. Corrected Manavgat label. Schoener's D , per-feature D and transfer on the original study areas are read from the source of Fig. 8, paper/labelfix_rerun/round5/out_official/figure_contrast_pairs.json. Collar transfer and ranks are from the source of Table S8, paper/labelfix_rerun/round5/collar/aoi_frame_transfer.csv, and the AoA shares from paper/labelfix_rerun/round5/collar/aoa_directed_pair_summary.csv. Every cell is checked against these files at build time (paper/figures/fig8_contrast_pairs.py).

	Manavgat and Muğla	Bejís and Montiferru
Schoener's D , mean 1-D	0.799 (highest)	0.479 (lowest)
per-feature D	0.74 to 0.87	0.23 to 0.77
transfer, frames as drawn	0.438, 0.345	0.594, 0.548
transfer, 10 km collar	0.493, 0.433	0.669, 0.624
rank of 20 on the collar, from the bottom	4th, 2nd	16th, 12th
target cells inside the AoA	0.876, 0.531	n/a

S3 Protocol detail and limitations

These protocols are given here in full. They are needed only to check the related results, not to follow the main argument. Sections 3.2 to 3.4, 3.7, 3.9, 3.10, 3.13 and 3.14 summarise them.

S3.1 Burned-area label, the reconstructed analysis grid, and the admissibility gate

Labels come from MODIS MCD64A1 Collection 6.1 (MODIS/061/MCD64A1), retrieved through Google Earth Engine (Gorelick et al. 2017). Its omission and commission errors (Boschetti et al. 2019) limit every model here. The product is exported onto the 30 m EPSG:4326 reference grid used by the rest of the pipeline. Each native ~500 m observation is therefore copied to the 30 m pixels below it.

The analysis grid is therefore **reconstructed, not native**. The 30 m grid is divided into non-overlapping blocks of $\text{round}(500 / 30) = 17 \times 17$ pixels, which gives a nominal cell edge of 510 m. Each cell is identified by its block indices `row_500m` and `col_500m`. This approximates the MODIS sinusoidal cell but does not reproduce it, because it is anchored to the Landsat/EPSG:4326 reference grid. Blocks at the edges are cut. The cell is square in degrees but not on the ground: about 510 m from north to south and 390 to 407 m from east to west. Its area is therefore 0.199 to 0.208 km², against 0.215 km² for the native MODIS cell (463 m edge). Every block size in this paper refers to the north-south dimension. The collar step of 0.45 km per cell (Section 3.12) lies between the two cell dimensions, so collar radii are nominal to within about 15 %.

The burn date of a cell is the **mode** of the burned sub-pixel day-of-year values in the block, and it is tested against the label window of the region. The exported raster has no burned pixels outside the window, so in this dataset one burned sub-pixel inside the window makes the cell burned. This can extend each scar by up to one cell at its edge, and it can mark a water cell as burned where the product does. The primary population keeps only cells dominated by natural vegetation, so water is excluded. The fraction of burned sub-pixels that agree with the modal date is recorded as `burn_date_pixel_agreement_fraction`, but no threshold is applied. The label never affects whether a cell is modelled: unburned, all-no-data and out-of-window cells all stay in the negative class.

Earlier burning is handled differently in each region; this is given in Section S3.1.1. The only historical exclusion applied by design removes the 2021 Muğla scar from the 2022 experiment of Section 3.14.

Before any modelling, each region passes a gate that asks one question: what fraction of the burned cells is dominated by natural vegetation? A region is admitted as a wildfire case when

this fraction is at least 0.50 and at least 30 burned cells are present. It is rejected as a cropland control when the cropland fraction is at least 0.50. The gate uses ESA WorldCover classes aggregated to the same cells. It separates burning of natural fuel from stubble burning after harvest, which MCD64A1 does not distinguish. The results are given in Section 4.1.

S3.1.1 Pre-label and historical burning, by region

Two kinds of earlier burning can affect a label. **Pre-label burning** is burning inside the predictor window of the region, which the pipeline can remove. **Historical burning** is burning in the five previous years, which no region screens for. Table S34 gives both, recomputed from the MCD64A1 export of each region (paper/code/ems_labels_1_prelabel.py, ems_labels_2_history.py; outputs paper/labelfix_rerun/labels/r1_prelabel.json, r2_history.json).

Table S34. Earlier burning by region, natural-vegetation population.

Region	Pre-label cells removed	Pre-label burned cells retained	Historical-burn cells (five years)	... of which burned again in the event
Manavgat 2021	none needed (no pre-label burning)	0	0	0
Bejís 2022	exclusion not applied	48 (49 in all valid cells; 10 burned again)	163	0
Muğla 2021	49	0	194	25
North Evia 2021	16	0	366	88
Montiferru 2021	61	0	58	32

Excluding the historically burned cells lowers the mean paired transfer contribution from +0.007 to +0.003 (at 10-cell blocking: 5 positive, 4 negative and 11 uncertain verdicts, against 6, 5 and 9). The within-region gain stays positive with interval support in every region: +0.062 (no change), +0.055, +0.088, +0.162 and +0.111 for Manavgat, Bejís, Muğla, Evia and Montiferru.

S3.1.2 The Manavgat label correction

The Manavgat 2021 label is corrected. The label first used for Manavgat was exported on 8 July 2026. This was before the month-alignment fix of the MCD64A1 query in the pipeline (commit 183be42, 11 July), and the export was not renewed. It therefore missed the first four days of the fire, 28 to 31 July. The error was an old export, not a code error. The corrected label only adds burned cells. No cell that burned under the original label becomes unburned, and no predictor or validity flag changes. In the primary population, the number of burned cells rises from 784 to 2,935. The Manavgat outputs were re-frozen with the pipeline at commit 6381f4c. Two small changes were made: a one-line patch to the window-closure diagnostic, and a runtime setting of the Earth Engine project (thermaltwin). Python 3.12.10 and scikit-learn 1.9.0 were used (versions in `ENVIRONMENT.md`). The correction, the re-freeze and the control run were carried out by the corresponding author with AI-assisted code (Section 3.13). They were not repeated by the author of the pipeline. This pipeline version writes one extra column, a flag for burning in earlier years, and it excludes no Manavgat cell. In a control run, the original label was re-frozen with the same code and environment. It reproduced the published outputs to within 10^{-5} . The exceptions are listed in the manifest: file paths, the column list, and one older pair that differs at the level of random-forest thread nondeterminism. Differences between the two runs are therefore caused by the label alone. The reproduction check of Section 3.13 covers the re-frozen outputs. The manifest, hashes and scripts are released in `paper/data/manavgat_2021/refreeze/`.

S3.1.3 Effect on Table 1

The Manavgat rows use the corrected label. It adds burned cells (784 to 2,935 in the primary population) and changes no predictor (Section 3.2). The rows become stronger, not weaker. Absolute AUCs rise by 0.04 to 0.11 across the three block sizes, and the gain holds at every

scale. The 1 km interval becomes narrower, from $[+0.055, +0.079]$ to $[+0.060, +0.073]$, because the region now has 814 blocks of 2 cells with burned cells instead of 235.

S3.2 Transferability diagnostics versus transfer

Twenty similarity measures from five families are each rank-correlated with the same quantity, the raw thermal transfer AUC over the twenty ordered directions, with one common pair-based bootstrap. The families are marginal predictor-distribution measures $P(x)$, burned-niche overlap $P(x|y=1)$, burned-pattern structure $P(y)$, and conditional feature-response direction $P(y|x)$.

The **marginal** family includes area-of-applicability dissimilarity in scaled, importance-weighted predictor space (Meyer and Pebesma 2021, 2022; Ludwig et al. 2023), climatic and geographic distance, and a domain classifier trained to separate source cells from target cells.

The **niche-overlap** family includes Schoener's D (Schoener 1968) and Warren's I (Warren et al. 2008) over the burned cells of each region. The **regime** family includes counts of burned patches and effective patch counts. The **conditional** family is the sign-agreement index: the fraction of predictors whose signed association points the same way in both regions, computed over all predictors and over the predictors with interval support.

Two properties of this design limit what it can show. First, the set of measures was recorded on 8 August 2026, together with its results under the original label (commit d0b20af in the authors' version history, available on request). It was not re-selected after the label correction. This is a record, not a pre-registration. Second, with five regions the effective sample is ten unordered pairs, and nineteen measures are evaluated at this power without family-wise error control. Where a measure passes an interval test but not a Bonferroni threshold, both results are reported.

S3.3 Same-geography event-to-event comparison (Muğla 2021 versus 2022)

Muğla is the only region in which a second fire is analysed on the same study area and grid. This allows the direction of feature-label associations to be compared with the place held fixed. The 2022 windows are set relative to the 2022 fire: a 58-day predictor window that closes the day before ignition, and a 49-day label window that opens on that day. These lengths are the same as in 2021.

This is not a clean design for transfer over time, and it is not presented as one. The 2022 fire started about six weeks earlier in the season than the 2021 fire, so year and seasonal phase are confounded, and no difference can be attributed to the year alone. The comparison is between two fires in one place. The pair is kept out of the twenty-direction matrix, and this is enforced in the released code.

Two further properties limit the comparison. A historical-burn exclusion applies only to the 2022 population: every cell that burned in 2021 is masked, so the scar of the previous year does not enter the analysis. This removes 3,073 cells, 2,941 of them in the primary population, and reduces the population from 41,730 to 38,790 rows (7.0 %). The two populations therefore share 38,789 of 38,790 cells and three identical static predictors. In the 2022-to-2021 direction, no burned target cell is in the source training data. Section S1.13 describes the consequences.

Table S19. Study regions, study areas and time windows. Bounding boxes are in EPSG:4326, as recorded. Window lengths in brackets are day counts, both ends included. The baseline years are the four years before each predictor window, with the same calendar dates.

Region	Bounding box (lon min, lat min, lon max, lat max)	Predictor window	Label window	Baseline years
Manavgat 2021 (Türkiye)	31.05, 36.72, 31.85, 37.35	2021-06-01 to 2021-07-27 (57 d)	2021-07-28 to 2021-08-31 (35 d)	2017, 2018, 2019, 2020

Region	Bounding box (lon min, lat min, lon max, lat max)	Predictor window	Label window	Baseline years
Bejís 2022 (Spain)	-1.05, 39.68, -0.35, 40.15	2022-06-15 to 2022-08-14 (61 d)	2022-08-15 to 2022-09-30 (47 d)	2018, 2019, 2020, 2021
Muğla 2021 (Türkiye)	27.1, 36.6, 28.9, 37.45	2021-06-01 to 2021-07-28 (58 d)	2021-07-29 to 2021-09-15 (49 d)	2017, 2018, 2019, 2020
North Evia 2021 (Greece)	23.05, 38.55, 23.85, 39.15	2021-06-05 to 2021-08-02 (59 d)	2021-08-03 to 2021-09-30 (59 d)	2017, 2018, 2019, 2020
Montiferru 2021 (Italy)	8.45, 40.05, 8.75, 40.27	2021-05-25 to 2021-07-23 (60 d)	2021-07-24 to 2021-08-31 (39 d)	2017, 2018, 2019, 2020

S3.4 Predictor provenance, compositing and the downscaler

Ten predictors are used, four baseline and six thermal. All are summarised per cell over the predictor window, through the chain shown in Fig. 2. All optical and thermal predictors come from Landsat 8 Collection 2 Level-2 (LANDSAT/LC08/C02/T1_L2). They are quality-screened per pixel with the QA_PIXEL band; the water bit is kept on purpose. The coarse thermal input is MODIS/061/MOD11A1.

- **NDVI** is the median of Landsat surface reflectance over the predictor window. It is the only baseline predictor that changes over time.
- **Elevation** and **slope** come from the Copernicus DEM GLO-30. Its heights are referenced to the EGM2008 geoid (EPSG:3855) in metres, according to the product handbook. The pipeline applies no datum conversion.
- **Land cover** is ESA WorldCover v200 (Zanaga et al. 2022), used as the dominant class code.
- **Current LST** is the median of Landsat surface temperature over the predictor window, in °C.
- **LST anomaly** is a z-score of the current LST median against the four baseline years. It is set to no-data where the baseline standard deviation is below 1.0 °C or there are too few observations.

- **TVDI and TVDI difference.** The Temperature-Vegetation Dryness Index (Sandholt et al. 2002) is computed in the LST-NDVI space. The wet and dry edges are the 2nd and 98th LST percentiles within each of twenty NDVI bins, and the index is limited to [0, 1]. The edges are percentiles of the LST values in a given scene, so they are fitted for each study area and window. A TVDI of 0.5 therefore means a different moisture state in each region. **TVDI difference** is the raw anomaly of this index against the same four baseline years: `tvdi_difference = current_tvdi - mean(baseline_tvdi)`. It is in index units, not standard deviations, so it is reported together with the z-scored channel.

- **Downscaled LST and fused LST.** A MODIS-to-Landsat downscaling model is trained on the predictor window and applied to the full 30 m grid. The fused product equals the observed Landsat LST where that is valid, and the downscaled surface only where it is not. Gap-filling therefore never replaces or blends a valid observation. The gap-filled share is 0.11 % to 9.70 % by region. The downscaler uses coordinates as inputs. This is the only way in which coordinate information enters a feature set from which Section 3.13 excludes coordinates. Section S1.7 gives the gain without these two channels.

The six thermal channels carry about two signals. In the primary population, current, fused and downscaled LST are correlated at 0.97 to 1.00 in every region. Current LST and TVDI are correlated at 0.88 to 0.98, and the two anomaly channels at 0.71 to 0.94. Two principal components of the six channels explain 92 % to 98 % of their variance, and their participation ratio is 1.6 to 1.8 (`paper/labelfix_rerun/labels/r4_dimensionality.json`).

Table S20. Input data. All retrieved through Google Earth Engine.

Input	Product and version	Native resolution	Use	Reference
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Input	Product and version	Native resolution	Use	Reference
Burned area	MODIS MCD64A1 Collection 6.1 (MODIS/061/MCD64A1)	500 m, monthly	label	(Giglio et al. 2018, 2021)
Surface reflectance and temperature	Landsat 8 Collection 2 Level-2 (LANDSAT/LC08/C02/T1_L2), QA_PIXEL screening, water bit preserved	30 m	NDVI, current LST, anomaly, TVDI	(U.S. Geological Survey Earth Resources Observation and Science (EROS) Center 2020)
Coarse LST	MODIS MOD11A1 v061 (MODIS/061/MOD11A1)	1 km, daily	downscaled and fused LST (gap-fill 0.11 % to 9.70 %)	(Wan et al. 2021)
Terrain	Copernicus DEM GLO-30 (EGM2008 heights)	30 m	elevation, slope	(European Space Agency 2022)
Land cover	ESA WorldCover v200 (2021)	10 m	dominant class, gate, population	(Zanaga et al. 2022)

Compositing uses the median over the predictor window. The anomaly channels refer to four baseline years (Table S19). WorldCover 2021 is later than four of the five fires (Section 3.4).

S3.5 Limitations, in full

Section 5.6 lists the limitations that affect the conclusions. All fourteen are given here.

(i) **No weather predictors** are used, so the behaviour of a combined thermal and weather predictor set is not known.

(ii) **The same-place comparison covers one region only**, and there, year and seasonal phase are confounded. This cannot be resolved in this study area. The two fires are 42 days apart in median burn day-of-year, neither year has a second fire at the phase of the other, and a calendar-matched comparison would have nine burned cells, against a gate minimum of thirty. The 2022 population also has only eleven blocks with burned cells, against the minimum of sixteen (Section S1.13). Its 331 burned cells do not resolve the thermal reversals at interval level, and the pair keeps the place fixed but not the population. The two Muğla populations were computed with the unmodified pipeline code and the same fixed environment, not read from a frozen export.

(iii)**All labels come from one burned-area product**, MCD64A1 (Giglio et al. 2018). Its omission and commission errors (Boschetti et al. 2019) limit every model here. Its accuracy in the Mediterranean was assessed against reference perimeters (Katagis and Gitas 2022). A second 500 m product, VIIRS VNP64A1 (Giglio 2024), covers both years but was not used as a label sensitivity.

(iv)**Evia has the most unusual class balance**, even on the extended study area. Its TSG prevalence is 0.287, the highest in the cohort, against 0.070 to 0.212 in the other four regions.

(v) **Each region contributes one fire season**, so regional concept shift is confounded with the weather of each fire. Labels from several years are needed to separate them.

(vi)**Point estimates depend on the software version**. Under scikit-learn 1.5.2 instead of 1.9.0, single-direction AUCs of the frame-transfer matrix move by up to 0.023 for the thermal model and 0.047 for the baseline, and paired differences by up to 0.045. The means move by at most 0.003. One direction changes interval support on each of the original and 10 km frames (paper/labelfix_rerun/round7/sklearn152/compare_vs_1_9_0.json). All reported numbers come from one verified version, and exact reproduction needs the archived environment.

(vii) **The low transfer into Manavgat is located but not explained**. Manavgat is the weakest target (0.435), has the largest matched shortfall (+0.319) and has the most unusual univariate profile in the cohort. Two processing causes were tested, and neither explains it. The quality screening of its coarse thermal input moves no signed association by more than 0.005 (Section S1.5), and its elevation reversal remains on

the collar (Section 4.4). Holding elevation reverses its raw LST association (Section S1.11). This rests on one region and one fire.

(viii) **Measures based on interval support are less stable than the point**

estimates. Five to six of the pipeline's interval bounds lie within 0.01 of 0.5, under both labels. One such flag, the NDVI of Manavgat, moved the supported-feature cosine on the original study areas from $p = 0.49$ to 0.70, when two equally valid bootstrap streams disagreed on it. The transfer verdicts are more stable. Across five bootstrap seeds, every verdict at 1 km blocking is stable, and the paired split of twelve positive, seven negative and one uncertain holds on every seed. At 5 km blocking, one level verdict and two paired-delta verdicts change with the seed. Exact counts of supported directions should be read at this precision.

(ix) **The five study areas are not comparable, and this cohort cannot fully correct this**

(Section 4.4). The collar results are reported together with the original results, not instead of them, because the collar radius is also a choice, and 5 km and 10 km do not agree exactly (0.591 against 0.589). The study areas were fixed earlier in the pipeline, in repo/, so they can be restricted but not extended. Montiferru, whose rectangle is already close to the fire, cannot be given distant cells. Their independence from the outcomes can also not be fully documented. Only the Montiferru rectangle follows a rule, and the Manavgat rectangle is dated before its first gate result. For Bejís and Muğla, the version history cannot show that the rectangle was fixed before the first gate result (Section 3.1). Future studies should fix the study area by a stated accessible-area rule (Barve et al. 2011) before any predictor is computed.

(x) **Other classifiers are compared by point estimate only.** The main results use a

random forest with unlimited depth, which fits local structure well and extrapolates

poorly. Section S1.8 shows that the transfer result does not depend on this choice.

Three other estimators, including a penalised linear model, range from 0.481 to 0.527

and have eleven to thirteen of twenty directions above chance. These are point

estimates without intervals, so the estimators are not ranked. Other estimator classes

were not tried.

(xi) **Burn timing and elevation are confounded in Manavgat.** The cells that burned in the first four days lie at a median of 219 m, against 512 m for the later burns and 1,004 m for unburned cells. The effects of timing and elevation cannot be separated in these data.

(xii) **The quality-screening comparison differs in code version as well as in screening.** The current pipeline step7 refuses the unscreened MODIS input, because the raster has no nodata tag and 8.1 % exact zeros. The unscreened version therefore uses the step7 of export time, and the screened version the current one. The two agree on elevation to four decimals and on every other signed AUC to within 0.005; the largest change is -0.0044 for downscaled LST. The within-region gain moves from $+0.067$ to $+0.068$. The confound could hide an effect only if two effects cancelled (Section S1.5).

(xiii) **The similarity tests rest on ten independent region pairs.** Five regions give twenty ordered directions, but the two directions of a pair share both regions and are not independent. All results of Section S1.22, positive and negative, have this low power. With ten pairs, a moderate true ordering would usually be missed.

(xiv) **The frame cost rests on few scars.** The full comparison of Section 4.3 is defined on seven scars in three regions. Its scar-level intervals are too narrow, because

1018 row A is repeated for each scar of a region. The region-level estimates (+0.137 over
1019 three regions, +0.160 over five) should be quoted.

1020 **S3.6 Leakage control and reproducibility, in full**

1021 A fixed set of forbidden columns is checked at every model fit. Coordinates (lon, lat, row,
1022 col and their normalised forms), every burn-date and label-provenance column, and the
1023 agreement fraction are excluded from all feature sets. The check is an assertion in the code.
1024 The natural-vegetation mask is used only to define the population, never as a predictor.
1025 Spatial blocking keeps neighbouring cells in the same fold.

1026 **Reproducibility.** All random processes use seed 42, and every bootstrap uses 1000 replicates.
1027 The transfer and adaptation analysis runs in an environment separate from the pipeline. Every
1028 within-region model was therefore refitted there and compared with the frozen pipeline
1029 output, and the independent adaptation code was compared with the pipeline's own. The
1030 within-region results agree exactly. The twenty CORAL transfers agree to within 1.3×10^{-8}
1031 with the re-frozen outputs (1.6×10^{-7} with the frozen ones), below the tolerance of 10^{-6} set in
1032 the repository (Section 3.13). All numbers were produced under scikit-learn 1.9.0 or checked
1033 against it. The version sensitivity is given in Section S3.5(vi).

1034 **Sensitivity analyses.** The within-region results are repeated for three block sizes, four
1035 classifier capacities and, in two regions, a second population. The transfer results are repeated
1036 for both feature sets, the CORAL sweep, the Evia study-area variant and three evaluation
1037 frames. Where a conclusion depends on one of these choices, the dependence is reported
1038 (Section S1).

S3.6.1 Resampling units

Uncertainty is estimated with a spatial-block bootstrap. Let $\beta_1, \dots, \beta_M$ be the blocks of Eq. (1) of the main text that contain cells of F . Replicate b draws M indices u_{bm} uniformly with replacement:

$$F^{*b} = \bigcup_{m=1}^M \beta_{u_{bm}}, \quad \text{CI}_{95} = [Q_{0.025}\{\theta(F^{*b})\}_b, Q_{0.975}\{\theta(F^{*b})\}_b]. \quad (S1)$$

Here θ is the statistic and Q gives the 2.5 and 97.5 percentiles over 1000 replicates, seed 42. A replicate with only one class is dropped. Differences are formed within each replicate, so they are paired. The block size of each interval is given with the result. Because blocks are resampled, the reliability of an interval depends on the number of blocks with at least one burned cell. These counts are reported with the intervals. Where a verdict rests on too few such blocks, it is reported as indicative.

The twenty transfer directions are not independent, because each region appears in eight of them. The **pair-cluster bootstrap** therefore resamples the ten unordered region pairs. Each pair carries its set π_p of ordered directions, and the carried values δ_{st} are averaged:

$$\bar{\delta}^{*b} = \frac{\sum_{m=1}^{10} \sum_{(s,t) \in \pi_{u_{bm}}} \delta_{st}}{\sum_{m=1}^{10} |\pi_{u_{bm}}|}. \quad (S2)$$

The interval has the same percentile form, with 1000 replicates for the direction-level intervals of Sections 4.4 and 4.5, and 20,000 for the unit comparison of Table S31. Clustering by target region replaces π_p by the four directions that share a target. A quantity with one value per held-out scar or target region gets a Student t interval over these n units:

$$\bar{x} \pm t_{0.975, n-1} s_x / \sqrt{n}. \quad (S3)$$

The effective sample is therefore ten pairs or five regions for direction-level intervals, and at most seven scars from three regions for scar-level intervals.

S3.6.2 Reproduction of the re-frozen outputs

The transfer analysis runs in an environment separate from the pipeline. The reproduction check of the repository therefore refitted every within-region model and all twenty CORAL transfers and compared them with the frozen pipeline output. For Manavgat this output is the one re-frozen on the corrected label (Section 3.2), produced with the pipeline at commit 6381f4c. Only two changes were made. A one-line patch lets the window-closure module accept a population column that the corrected label adds; its diff is released with the re-freeze. A runtime setting points the Earth Engine calls at the project `thermaltwin`, which changes only the infrastructure. The within-region results agree exactly, and the transfer directions to within 1.3×10^{-8} , below the repository tolerance of 10^{-6} . This tolerance applies to the CORAL and within-region check. The frame-transfer script of Section 4.4 fits its forests in parallel, so its values vary between runs by up to 2×10^{-6} (1.4×10^{-6} in a rerun on 29 September 2026, `paper/labelfix_rerun/round7/`). This is within the tolerance of 10^{-5} applied to it, and the values are stable at the printed precision. The effect of an unpinned library version is given in Section S3.5(vi).

S3.7 Transfer-gap decomposition and the concept-shift diagnostic, in full

For each direction, the gap between the within-region skill of the target and the raw transfer result is split in two parts. One part is what the best label-free adaptation recovers, and the other is what it does not recover. The recovered fraction is $(\text{adapted} - \text{raw}) / (\text{within} - \text{raw})$, signed and not clipped, with its interval from the same paired bootstrap. It is an upper limit on what a correction of the predictor distributions can achieve. Section 4.3 shows that the

remainder should not be read as a concept-shift residual, because much of it already appears inside a single region (Section S1.10).

Concept shift is examined with the **signed univariate association**. For each numeric predictor, the raw ROC-AUC of the predictor against burned is computed in each region and is not folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 shows a direction, not weakness. A reversal is called bootstrap-supported only when the point estimates of the two regions lie on opposite sides of 0.5 **and the interval of each region excludes 0.5**, under the same 10-cell spatial-block bootstrap. This is stricter than requiring disjoint intervals. A feature with disjoint intervals, one of which includes 0.5, has not been shown to point in any direction in that region, so it is recorded as a point reversal only. Section S2 repeats the rule next to the counts, and `conditional_similarity_transfer.json` stores it as metadata.

S3.7.1 Degenerate replicates

Replicates with $|G| < 10^{-6}$ are dropped.

S3.8 Label-blind adaptation, full specification

Region-wise z-score. The numeric features of each region are standardised with the statistics of that region: source statistics from source data and target statistics from target data, never pooled. For feature j in region R ,

$$z_{ij} = \frac{x_{ij} - \mu_j^R}{\sigma_j^R}, \quad \mu_j^R = \frac{1}{n_j^R} \sum_{i \in O_j^R} x_{ij}, \quad \sigma_j^R = \left(\frac{1}{n_j^R} \sum_{i \in O_j^R} (x_{ij} - \mu_j^R)^2 \right)^{1/2}, \quad (S4)$$

where O_j^R contains the n_j^R cells with an observed value (ddof 0). A missing value is first set to μ_j^R , so it becomes zero, and $\sigma_j^R < 10^{-12}$ is replaced by 1. Land cover is not transformed. Target feature statistics, but never target labels, are therefore used in the adapted models. The

1103 classifier is refitted on the z-scored source and applied to the z-scored target. This removes
 1104 differences in the mean and variance of each feature.

1105 **CORAL after region-wise z-score.** The source covariance is aligned to the target covariance
 1106 with the standard whitening and recolouring map (Sun et al. 2016). With Z_R the $n_R \times d$
 1107 matrix of z-scored numeric features, one row per cell,

$$1108 \quad Z_s^{\text{al}} = Z_s (C_s + \lambda I)^{-1/2} (C_t + \lambda I)^{1/2}, \quad C_R = \frac{1}{n_R} \sum_{i=1}^{n_R} (z_i - \bar{z}_R)(z_i - \bar{z}_R)^\top, \quad (S5)$$

1109 with $\lambda = 10^{-5}$ (ddof 0). Both means are zero after Eq. (S4), so the mean terms of the general
 1110 map are zero. Matrix powers use a symmetric eigendecomposition, with eigenvalues set to at
 1111 least 10^{-12} . **The transform is applied to the source only.** The target stays at Z_t , and the
 1112 classifier is refitted on Z_s^{al} . Neither method uses a target label, and this is checked at run time.
 1113 The sensitivity to λ was tested over nine values on four of the twenty directions; transfer AUC
 1114 moved by at most 0.014, and λ was not selected on performance. The $\lambda = 1$ of the original
 1115 CORAL formulation lies outside this range. The value used throughout is $\lambda = 10^{-5}$ (Section
 1116 S1.2).

1117 **S5 Target-label recovery curve**

1118 **S5.1 Purpose**

1119 The main text shows that label-free alignment does not close the transfer gap. This section
 1120 asks how much of the gap a small number of target labels can close. It is not a proposed
 1121 method or an operational claim, because the labels come from the fire that is predicted. It
 1122 covers three of the five regions (Manavgat, Bejís and Muğla) and all six directions among
 1123 them. Source: the pipeline’s few-shot recovery diagnostic on the corrected label, analysis

7348dfe7..., released in paper/labelfix_rerun/exports/few_shot_recovery_7348dfe7/
(recovery_curve.csv, repeat_metrics.csv, report.md); every value below is read from it.

S5.2 Design

The unit of labelling effort is one 10-cell (≈ 5 km) spatial block of the target. For a budget of k blocks, k target blocks are drawn from the training folds of a 5-fold spatially blocked split of the target. Their labels are added to the source training set, and the refitted model is scored on the held-out target folds. The budgets are 0, 1, 2, 4, 8, 16 and 32 blocks, with ten repeated draws for every $k > 0$. Budget 0 is raw transfer. The **ceiling** is a target-only model at the same 10-cell blocking (0.777 for Muğla, 0.824 for Bejís and 0.882 for Manavgat). It reproduces the pipeline's own large-block values exactly. The **recovered fraction** is $(\text{few-shot} - \text{raw}) / (\text{ceiling} - \text{raw})$, the share of the gap between raw transfer and the ceiling that the labels close; it is signed and not clipped. The intervals are **selection intervals**: the 2.5th and 97.5th percentiles over the ten block draws. They describe only which blocks were drawn, and no hypothesis test is made.

S5.3 Result

Table S22. Few-shot recovery of target ROC-AUC, thermal model, natural-vegetation population. Raw = source-only transfer (budget 0); ceiling = target-only model at the same 10-cell blocking. Values are means over ten repeated block draws.

Direction	Raw	1 blk	2 blk	4 blk	8 blk	16 blk	32 blk	Ceiling	Gap closed at 32
Bejís → Manavgat	0.314	0.518	0.565	0.659	0.698	0.750	0.820	0.882	89 %
Muğla → Bejís	0.583	0.578	0.603	0.625	0.666	0.743	0.789	0.824	85 %
Manavgat → Bejís	0.396	0.447	0.456	0.541	0.598	0.690	0.752	0.824	83 %
Muğla → Manavgat	0.345	0.358	0.380	0.407	0.437	0.505	0.624	0.882	52 %
Manavgat → Muğla	0.438	0.447	0.453	0.468	0.490	0.532	0.571	0.777	39 %
Bejís → Muğla	0.618	0.576	0.577	0.578	0.598	0.637	0.666	0.777	30 %

1142 **At 32 blocks, three of six directions close 83 to 89 % of the gap to the ceiling.** Two of
1143 these three started below chance. The other three close 30 to 52 %. Thirty-two blocks contain
1144 2,740 to 2,970 labelled cells, 7 to 20 % of the natural-vegetation population of the target. This
1145 is not a small budget. For Bejís, the 32 labelled blocks contain on average 880 of its 1,100
1146 burned cells, and for Manavgat about 1,800 of 2,935.

1147 **Recovery is slow where transfer is worst.** Muğla → Manavgat has the largest gap between
1148 raw transfer and the ceiling. It is still below 0.5 after 8 labelled blocks and closes 52 % at 32.

1149 **Small budgets harm the direction that already transfers.** Bejís → Muğla is above chance
1150 without labels (0.618), and target labels help it least. The curve is below the raw value at 1, 2,
1151 4 and 8 blocks (0.576 to 0.598), rises above it only at 16 blocks, and closes 30 % at 32.

1152 **S5.4 Limits**

1153 1. **Three regions, six directions.** Evia and Montiferru are not included, so this covers six
1154 of the twenty directions and cannot cover all five regions. Six directions cannot
1155 support a general label budget, so none is given.

1156 2. **It needs labelled target cells** from the fire that is predicted. Nothing here is a label-
1157 free method, and nothing here changes the negative result for label-free alignment.
1158 Labels from earlier fires in the same region were not tested.

1159 3. **The interval is a selection interval. At the largest budgets it becomes narrow**
1160 **because the pool of blocks is used up, not because the value is well estimated.** Per
1161 outer fold, the target has on average about 12 blocks with both classes in Bejís, 29 in
1162 Manavgat and 48 in Muğla. At 16 and 32 blocks, the draws therefore use almost all of
1163 these blocks and select almost the same set (`direction_budget_feasibility.csv`).

1164 4. **The ceiling is the 10-cell target-only value**, not the ≈ 1 km within-region value of
1165 Table 1, so it is lower. The recovered fractions can only be read against this matched

1166 ceiling, and on the original study areas, which are not comparable across regions
 1167 (Section 4.4).

1168 S6 Code

1169 The analysis code is public at <https://github.com/dryuemco/thermal-twin> under the MIT
 1170 licence. This section is a guide to it; the README of the repository gives the same
 1171 information. The satellite processing pipeline is a separate public repository,
 1172 <https://github.com/emrehann17/satellite-thermal-digital-twin> (MIT licence). The re-frozen
 1173 Manavgat outputs were produced at its commit 6381f4c. The outputs of the other regions
 1174 were produced at earlier commits and are reproduced by it (Section S3.6.2).

Path	Role
<code>paper/code/</code>	Analysis and verification scripts. <code>_canonical.py</code> loads every modelling dataset, verifies its SHA-256 and asserts the leakage exclusions of Section 3.13 at every fit.
<code>paper/code/appendix_tables.py</code>	Rebuilds Tables S2 to S7 and S9 to S18 row by row from source files whose SHA-256 values are pinned in <code>paper/labelfix_rerun/round5/tables/SOURCES.sha256</code> , and compares them with this document.
<code>paper/figures/</code>	One script per figure. Each asserts every plotted value against the frozen outputs and the manuscript text, and checks its layout.
<code>paper/figures/check_all.py</code>	Runs all figure scripts and <code>appendix_tables.py</code> , restores the committed outputs byte for byte, and exits 0 when everything passes.
<code>paper/tex/</code>	Builds the manuscript and this document from their Markdown sources and checks the port: every number survives, and every cross-reference, citation and supplementary reference resolves.
<code>paper/labelfix_rerun/</code>	The corrected-label outputs from which every corrected number is read: <code>code/</code> , <code>geometry/</code> , <code>inference/</code> , <code>labels/</code> and <code>round3/</code> to <code>round7/</code> hold the analysis outputs, <code>pipeline/</code> the re-frozen pipeline outputs for Manavgat, and <code>exports/</code> the released pipeline diagnostics (few-shot recovery, CORAL λ sweep, window closure), each with its SHA-256 in <code>exports/SHA256SUMS.txt</code> .

1175 All commands are run from the repository root. The environment is Python 3.12.10 with
 1176 NumPy 2.4.4, pandas 3.0.2 and scikit-learn 1.9.0, fixed in `ENVIRONMENT.md`. Point estimates
 1177 depend on the scikit-learn version (Section S3.5(vi)). Fig. 1 also needs cartopy; without it,
 1178 `check_all.py` reports this figure as skipped. Tables S1 and S20 to S34 and the values in the

1179 text are not checked by `appendix_tables.py`; they name their source files in their captions or
1180 text.

1181 **S7 Data**

1182 All satellite inputs are public and were retrieved through Google Earth Engine (Table S20).

1183 No proprietary or restricted data were used.

1184 The five modelling datasets are in the analysis repository, one per region, as
1185 `paper/data/<region>/step8a_500m_modeling_dataset.parquet`. Each is checked on load
1186 against the SHA-256 recorded in `paper/code/_canonical.py`. For Bejís, Muğla, North Evia
1187 and Montiferru this hash equals the pipeline’s own record. For Manavgat the file is the
1188 corrected-label re-freeze (SHA-256 5a5e876c...; Section S3.1.2). The pipeline’s record still
1189 names the original-label table (SHA-256 054a1961...), which is not in the repository and can
1190 be regenerated by the pipeline. The frozen numeric outputs behind every table and figure are
1191 in the same repository, identified by the paths and hashes given in the table captions and in
1192 `SOURCES.sha256`. The data derived from Landsat, MODIS, the Copernicus DEM and ESA
1193 WorldCover remain subject to the terms of these products, which are given in the
1194 Declarations.

1195 **References**

1196 Barve N, Barve V, Jiménez-Valverde A, et al (2011) The crucial role of the accessible area in
1197 ecological niche modeling and species distribution modeling. *Ecological Modelling*
1198 222:1810–1819. <https://doi.org/10.1016/j.ecolmodel.2011.02.011>

1199 Boschetti L, Roy DP, Giglio L, et al (2019) Global validation of the Collection 6 MODIS
 1200 burned area product. *Remote Sensing of Environment* 235:111490.
 1201 <https://doi.org/10.1016/j.rse.2019.111490>

1202 European Space Agency (2022) Copernicus DEM GLO-30 [dataset]. European Space
 1203 Agency. <https://doi.org/10.5270/ESA-c5d3d65>

1204 Giglio L (2024) VIIRS/NPP burned area monthly L4 global 500m SIN grid V002 [dataset].
 1205 NASA Land Processes Distributed Active Archive Center.
 1206 <https://doi.org/10.5067/VIIRS/VNP64A1.002>

1207 Giglio L, Boschetti L, Roy DP, et al (2018) The Collection 6 MODIS burned area mapping
 1208 algorithm and product. *Remote Sensing of Environment* 217:72–85.
 1209 <https://doi.org/10.1016/j.rse.2018.08.005>

1210 Giglio L, Justice C, Boschetti L, Roy D (2021) MODIS/Terra+Aqua burned area monthly L3
 1211 global 500m SIN grid V061 [dataset]. NASA Land Processes Distributed Active Archive
 1212 Center. <https://doi.org/10.5067/MODIS/MCD64A1.061>

1213 Gorelick N, Hancher M, Dixon M, et al (2017) Google Earth Engine: Planetary-scale
 1214 geospatial analysis for everyone. *Remote Sensing of Environment* 202:18–27.
 1215 <https://doi.org/10.1016/j.rse.2017.06.031>

1216 Katagis T, Gitas IZ (2022) Assessing the accuracy of MODIS MCD64A1 C6 and FireCCI51
 1217 burned area products in Mediterranean ecosystems. *Remote Sensing* 14:602.
 1218 <https://doi.org/10.3390/rs14030602>

1219 Ludwig M, Moreno-Martinez A, Hölzel N, et al (2023) Assessing and improving the
 1220 transferability of current global spatial prediction models. *Global Ecology and Biogeography*
 1221 32:356–368. <https://doi.org/10.1111/geb.13635>

1222 Meyer H, Pebesma E (2021) Predicting into unknown space? Estimating the area of
 1223 applicability of spatial prediction models. *Methods in Ecology and Evolution* 12:1620–1633.
 1224 <https://doi.org/10.1111/2041-210X.13650>

1225 Meyer H, Pebesma E (2022) Machine learning-based global maps of ecological variables and
 1226 the challenge of assessing them. *Nature Communications* 13:2208.
 1227 <https://doi.org/10.1038/s41467-022-29838-9>

1228 Sandholt I, Rasmussen K, Andersen J (2002) A simple interpretation of the surface
 1229 temperature/vegetation index space for assessment of surface moisture status. *Remote Sensing*
 1230 *of Environment* 79:213–224. [https://doi.org/10.1016/S0034-4257\(01\)00274-7](https://doi.org/10.1016/S0034-4257(01)00274-7)

1231 Schoener TW (1968) The Anolis lizards of Bimini: Resource partitioning in a complex fauna.
 1232 *Ecology* 49:704–726. <https://doi.org/10.2307/1935534>

1233 Sun B, Feng J, Saenko K (2016) [Return of frustratingly easy domain adaptation](#). In:
 1234 *Proceedings of the thirtieth AAAI conference on artificial intelligence (AAAI-16)*. AAAI
 1235 Press, Phoenix, Arizona, USA, pp 2058–2065

1236 U.S. Geological Survey Earth Resources Observation and Science (EROS) Center (2020)
 1237 Landsat 8-9 operational land imager / thermal infrared sensor level-2, collection 2 [dataset].
 1238 US Geological Survey. <https://doi.org/10.5066/P9OGBGM6>

1239 Wan Z, Hook S, Hulley G (2021) MODIS/Terra land surface temperature/emissivity daily L3
 1240 global 1km SIN grid V061 [dataset]. NASA Land Processes Distributed Active Archive
 1241 Center. <https://doi.org/10.5067/MODIS/MOD11A1.061>

1242 Warren DL, Glor RE, Turelli M (2008) Environmental niche equivalency versus
 1243 conservatism: Quantitative approaches to niche evolution. *Evolution* 62:2868–2883.
 1244 <https://doi.org/10.1111/j.1558-5646.2008.00482.x>

- 1245 Zanaga D, Van De Kerchove R, Daems D, et al (2022) ESA WorldCover 10 m 2021 v200.
- 1246 Zenodo. <https://doi.org/10.5281/zenodo.7254221>